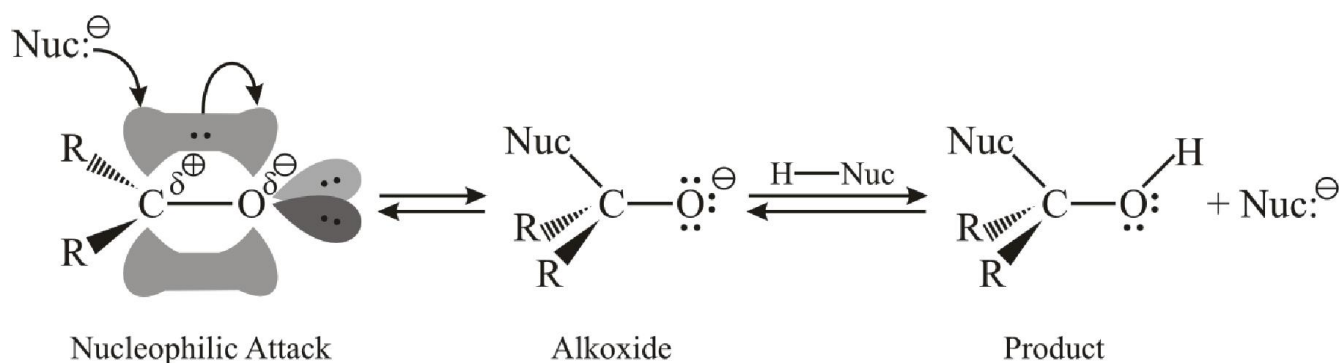


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ALDEHYDES & KETONES

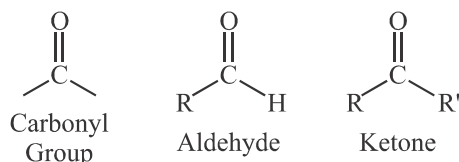
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THEORY

1. INTRODUCTION

Aldehydes and Ketones are organic compounds belonging to the class of **carbonyl compounds**. Aldehydes have a general formula of $RCHO$ and ketones have a general formula of $RCOR'$. Carbonyl compounds have a common group $C=O$.



In chemical industry ketones and aldehydes are used as solvents, starting materials, and reagents for the synthesis of other products.

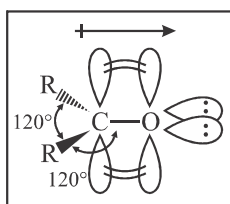
Formaldehyde is well known as the **formalin** solution used to preserve biological specimens and is also used in formation of several important polymers such as **Bakelite**. Ketones dissolve a wide range of organic materials having convenient boiling points for easy distillation, and have low toxicities.

Many other ketones and aldehydes are used as flavourings and additives to foods, drugs and other products.

Benzaldehyde is the primary component of almond extract, and (–) carvone gives spearmint chewing gum its minty flavour.

2. PHYSICAL PROPERTIES

The carbonyl carbon atom is sp^2 hybridized and bonded to three other atoms through coplanar sigma bonds oriented about 120° apart. The unhybridized p orbital overlaps with a p orbital of oxygen to form a pi bond.

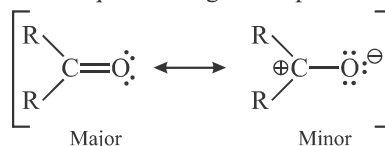


Functional Group	Length	Energy
Ketone $C=O$ bond	1.23 Å	745 kJ/mol (178 kcal/mol)
Alkene $C=C$ bond	1.34 Å	611 kJ/mol (146 kcal/mol)

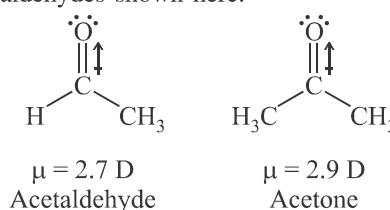
2.1 Dipole Moment

The double bond of the carbonyl group has a large dipole moment because oxygen is more electronegative than carbon, and the bonding electrons are not shared equally. In particular, the less tightly held pi electrons are pulled more strongly toward the

oxygen atom, giving ketones and aldehydes larger dipole moments than most alkyl halides and ethers. We can use resonance forms to symbolize this unequal sharing of the pi electrons.



The first resonance form is more important because it involves more bonds and less charge separation. The contribution of the second structure is evidenced by the large dipole moments of the ketones and aldehydes shown here.

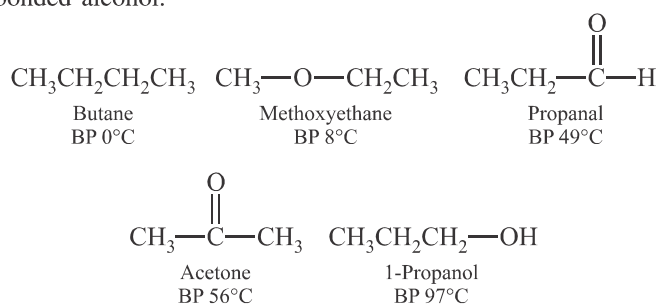


The positively polarized carbon atom acts as an electrophile (Lewis acid), and the negatively polarized oxygen acts as a nucleophile (Lewis base).

2.2 Boiling Point

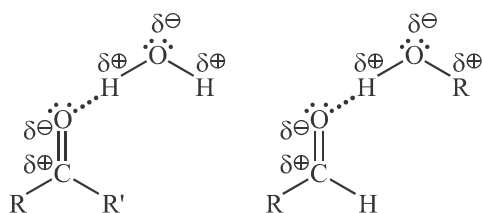
Ketones and aldehydes have no $O-H$ or $N-H$ bonds, however, so their molecules cannot form hydrogen bonds with each other. Their boiling points are therefore lower than those of alcohols of similar molecular weight.

The ketone and the aldehyde are more polar and higher-boiling than the ether and the alkane, but lower-boiling than the hydrogen-bonded alcohol.



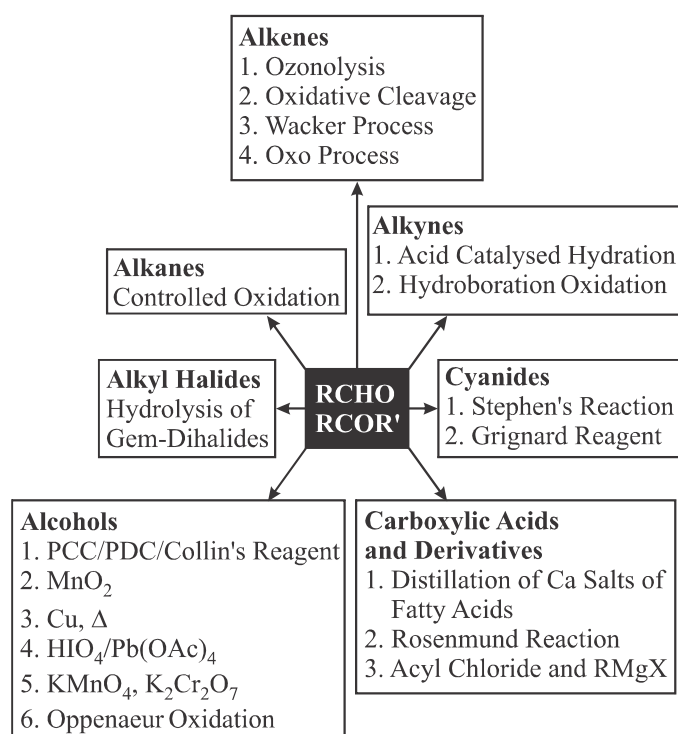
2.3 Solubility

Although pure ketones and aldehydes cannot engage in hydrogen bonding with each other, they have lone pairs of electrons and can act as hydrogen bond acceptors with other compounds having $O-H$ or $N-H$ bonds. For example, the $-OH$ hydrogen of water or an alcohol can form a hydrogen bond with the unshared electrons on a carbonyl oxygen atom.



Because of this hydrogen bonding, ketones and aldehydes are good solvents for polar hydroxylic substances such as alcohols. They are also remarkably soluble in water. Acetaldehyde and acetone are miscible (soluble in all proportions) with water. Other ketones and aldehydes with up to four carbon atoms are fairly soluble in water. These solubility properties are similar to those of ethers and alcohols, which also engage in hydrogen bonding with water.

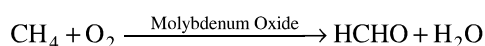
3. PREPARATION OF CARBONYL COMPOUNDS



3.1 Alkanes

Controlled oxidation of alkanes under high temperature and pressure in the presence of a catalyst gives carbonyl compounds.

Example - 1

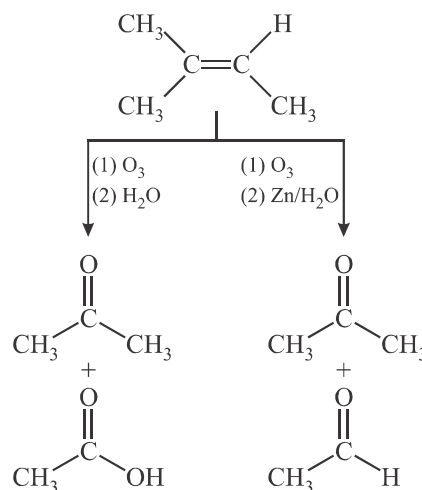


3.2 Alkenes

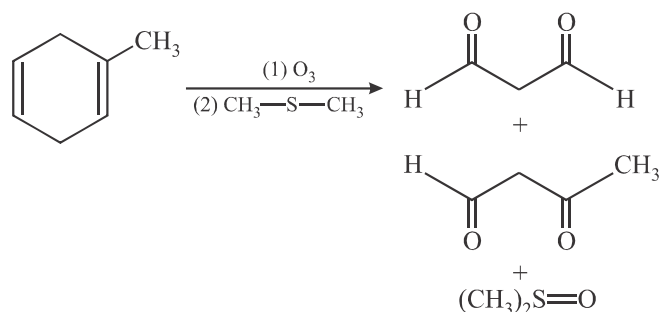
3.2.1 Ozonolysis

Alkenes can be oxidized to aldehydes and ketones if reacted with O_3 and then reduced with Zn or dimethyl sulphide, $(\text{CH}_3)_2\text{S}$. If a reducing agent is not used, the aldehyde formed cannot be isolated as it is further oxidized to carboxylic acids.

Example - 2



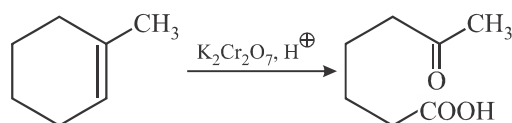
Example - 3



3.2.2 Oxidative Cleavage

Strong oxidizing agents such as acidic $\text{K}_2\text{Cr}_2\text{O}_7$ or alkaline KMnO_4 cause oxidative cleavage of double bond to form ketones and carboxylic acids.

Example - 4

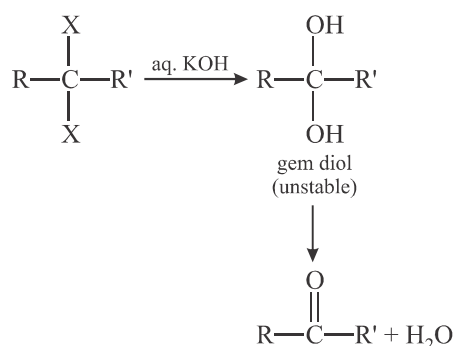


Note... 1. Formation of aldehydes from terminal alkynes requires the use of sterically hindered alkyl borane reagent such as Sia_2BH otherwise the reaction will not terminate at the vinyl borane stage and yield of aldehyde is not satisfactory.

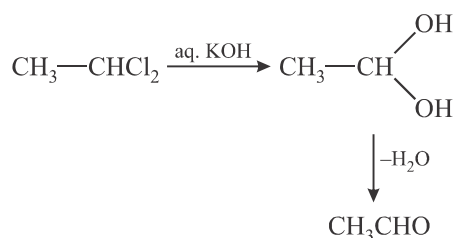
- Hydration of an internal alkyne always yields a ketone.
- Acid catalyzed hydration of acetylene yields acetaldehyde. All other terminal alkynes yield ketone.
- Hydroboration oxidation of terminal alkynes always yields an aldehyde.

3.4 Alkyl Halides

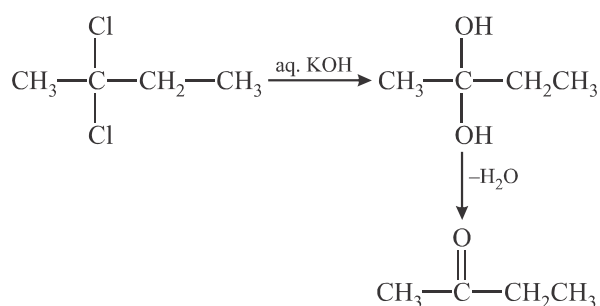
Hydrolysis of geminal dihalides yields carbonyl compounds.



Example - 12

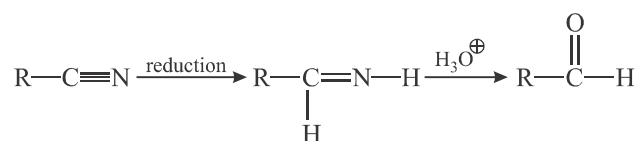


Example - 13



3.5 Alkyl Cyanides

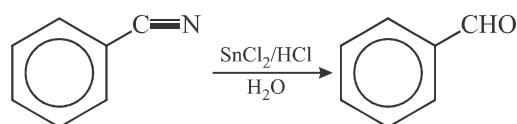
3.5.1 Reduction of Cyanides



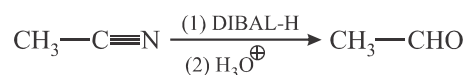
Note... 1. If reduction is carried out with SnCl_2/HCl , the reaction is called **Stephen's Reduction**.

- The reduction can also be carried out with **DIBAL-H** (Di-Isobutyl Aluminium Hydride).
- Only aldehydes are prepared by this method.

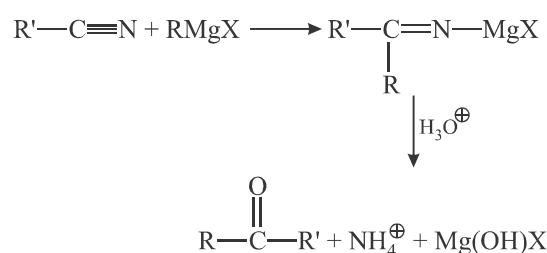
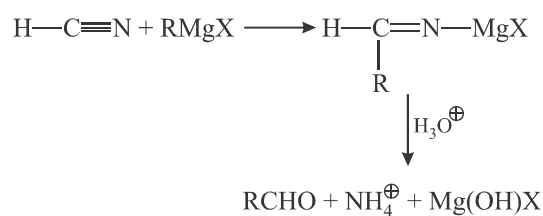
Example - 14



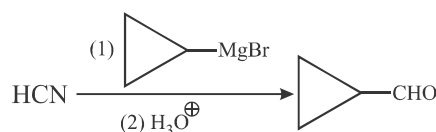
Example - 15

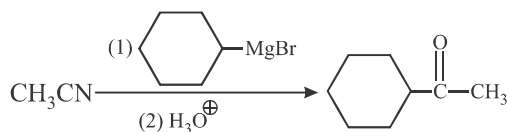


3.5.2 Grignard Reagent



Example - 16



Example - 17


Note...

1. Reaction of Grignard reagent with HCN gives aldehyde.
2. Reaction of Grignard reagent with alkyl cyanide (RCN) gives ketone.

3.6 Alcohols

Primary alcohols (1°) are oxidized to aldehydes whereas secondary alcohols (2°) are oxidized to ketones using suitable reagents. Tertiary alcohols (3°) cannot be oxidized using following methods.

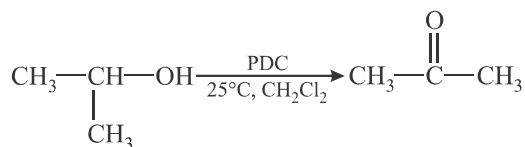
3.6.1 PCC/PDC/Collin's Reagent

Pyridinium chlorochromate (PCC) and Pyridinium dichromate (PDC) oxidize 1° alcohols and 2° alcohols to aldehydes and ketones respectively.

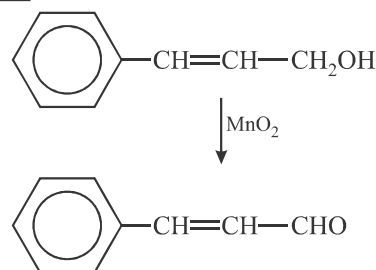
Note...

1. Collin's Reagent-Dipyridine Chromium (VI) oxide. $\text{CrO}_3 \cdot 2\text{C}_2\text{H}_5\text{N}$ in CH_2Cl_2 .
2. Corey's Reagent - PCC in CH_2Cl_2 .
3. Cr (VI) oxidizes 1° alcohols to aldehydes in anhydrous medium and to carboxylic acids in aqueous medium.

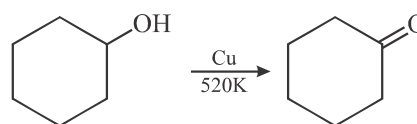
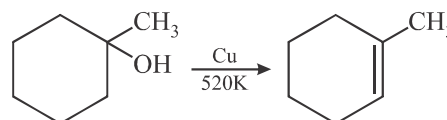
Example - 18

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3.6.2 MnO₂

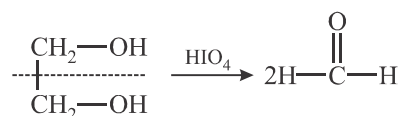
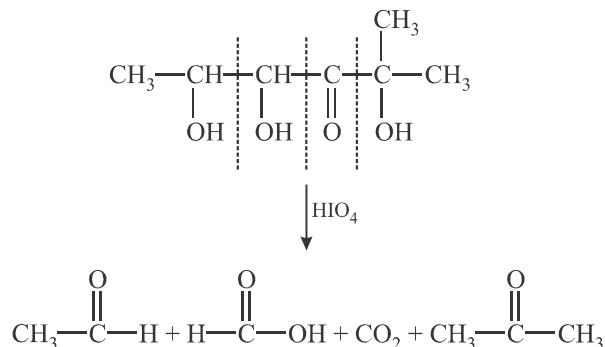
Allylic, benzylic and propargylic alcohol are oxidized to aldehydes and ketones by MnO_2 .

Example - 20

3.6.3 Cu, Δ - Vapour Phase Dehydrogenation

Vapours of alcohol are passed over heated copper to convert them into aldehydes and ketones. 3° alcohols are oxidized to alkenes.

Example - 21

Example - 22

3.6.4 Periodic Acid/Lead Tetraacetate

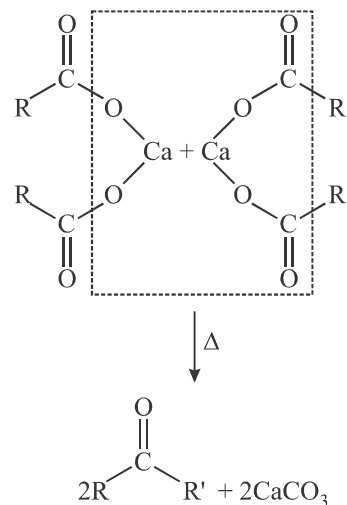
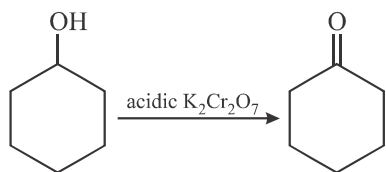
Oxidation of vicinal diols can be carried out with oxidizing agents such as **periodic acid**, HIO_4 and **lead tetra acetate**, $(\text{CH}_3\text{COO})_4\text{Pb}$. These oxidizing agents also cleave vicinal diketo compounds to carboxylic acids.

Example - 23

Example - 24


3.6.5 KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, Jones Reagent

Oxidizing agents such as alkaline KMnO_4 , acidic $\text{K}_2\text{Cr}_2\text{O}_7$ and Jones Reagent ($\text{CrO}_3 + \text{H}_2\text{O}/\text{Acetone}$) oxidize secondary alcohols to ketones. With primary alcohols, they yield carboxylic acids.

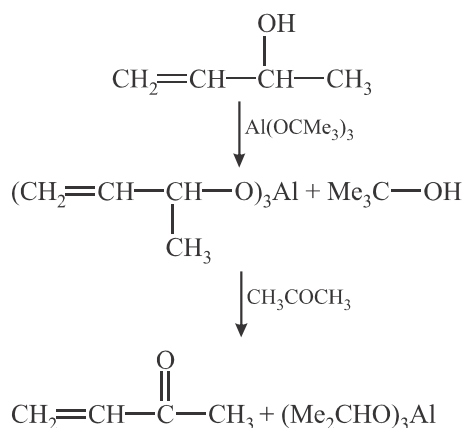
Example - 25



3.6.6 Oppenauer Oxidation

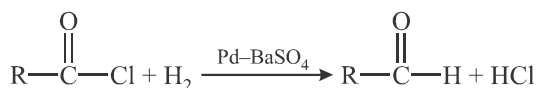
Secondary Alcohols are oxidized to ketones by refluxing them with aluminium tert-butoxide and then acetone is added. The reaction is selective and is particularly used to oxidize unsaturated secondary alcohols.

Example - 26

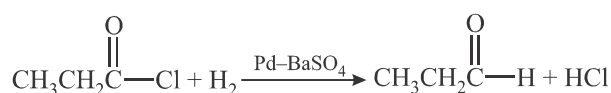


3.7.2 Rosenmund's Reduction

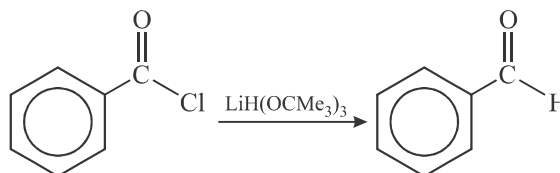
Acid chloride is reduced in boiling xylene using **palladium catalyst supported on barium sulphate**. Small amounts of quinoline and sulphur are also added to increase poisoning. **Lithium tri-tert-butoxyaluminium hydride** can also reduce acid chloride to aldehyde.



Example - 27



Example - 28

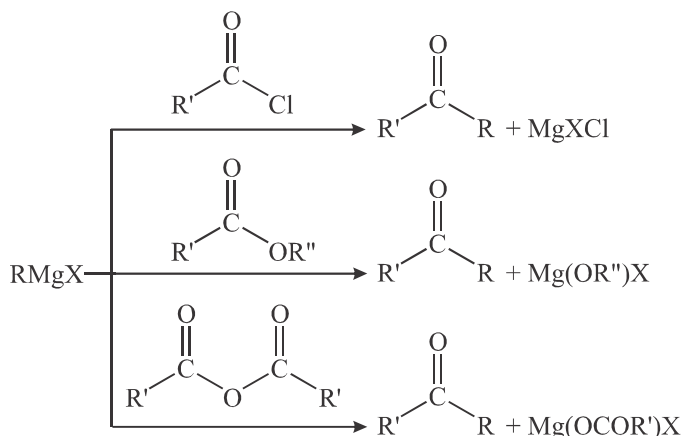


3.7 Carboxylic Acids and Derivatives

3.7.1 Decarboxylation of Ca Salts

A calcium salt of monocarboxylic acids on heating gives ketone. If heated with calcium salt of formic acid, the product is an aldehyde.

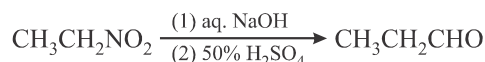
3.7.3 Acid Derivatives and Grignard Reagent



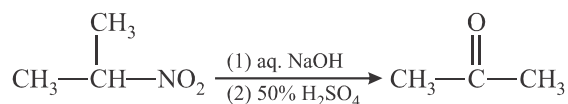
3.8 Nitro Alkanes – NEF Carbonyl Synthesis

Nitro compounds dissolve in aqueous sodium hydroxide to form salts. When this salt is acidified with 50% H_2SO_4 , aldehydes (from 1°) and ketones (from 2°) are formed.

Example - 29



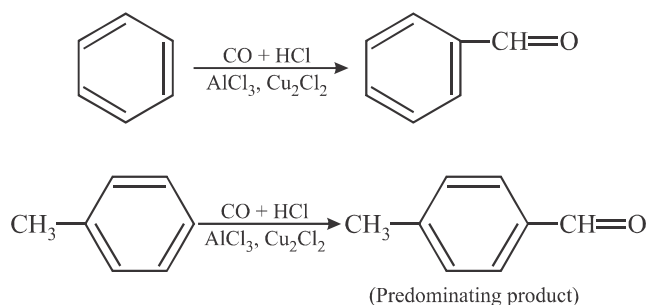
Example - 30



3.9 Aromatic Aldehydes

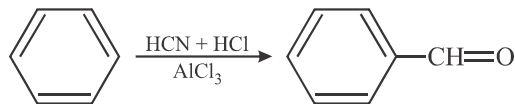
3.9.1 Gattermann Koch Aldehyde Synthesis

Carbon monoxide and HCl are passed through benzene solution in ether in the presence of AlCl_3 (catalyst) to produce benzaldehyde.



3.9.2 Gattermann Aldehyde Synthesis

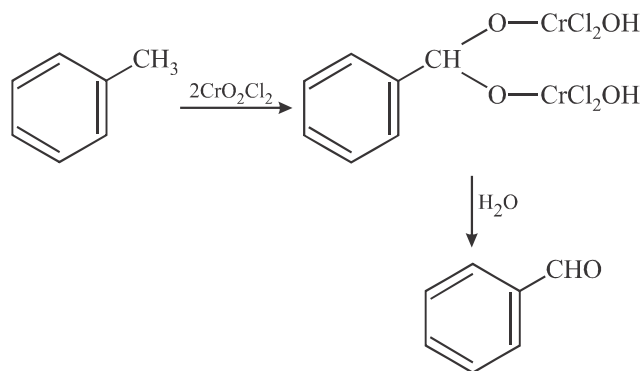
Benzene on treatment with hydrogen cyanide and hydrogen chloride in the presence of aluminium chloride initially gives imine which is hydrolysed into benzaldehyde.



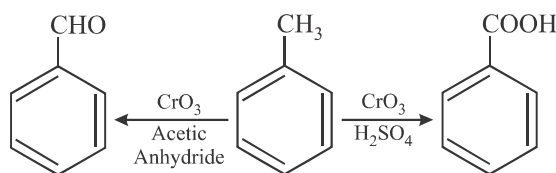
3.9.3 Etard Reaction

Toluene on treatment with chromyl chloride in CCl_4 , and on following hydrolysis gives benzaldehyde.

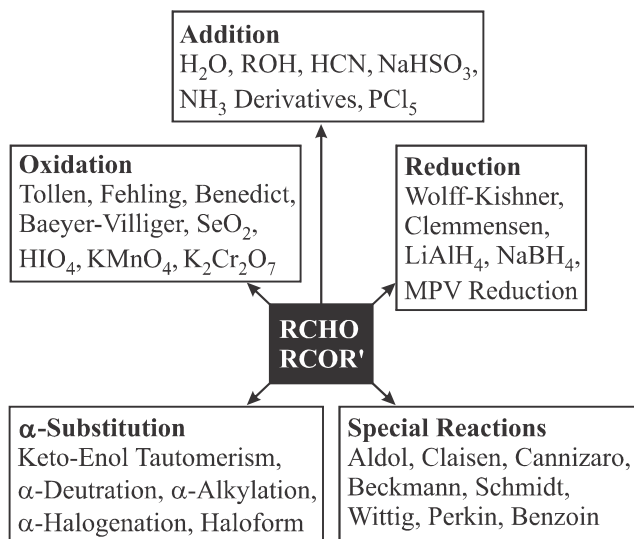
Toluene on treatment with CrO_3 in acetic anhydride also gives benzaldehyde. The formation of benzalidene acetate as intermediate prevents further oxidation to acid.



Example - 31



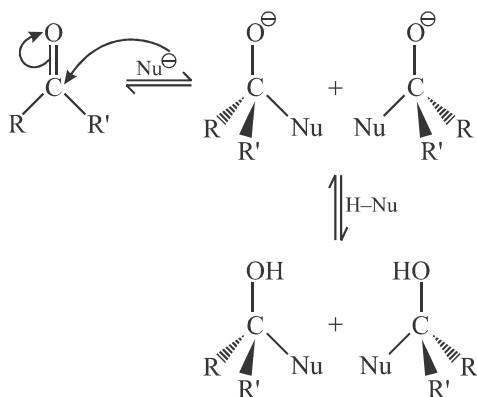
4. REACTIONS OF CARBONYL COMPOUNDS



4.1 Addition Reactions

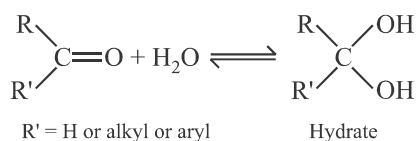
4.1.1 General Mechanism

Trigonal carbonyl compound gets converted into a tetrahedral product. Electronic as well as steric factor suggest that aldehyde is more reactive in comparison to ketone. Racemisation may proceed.



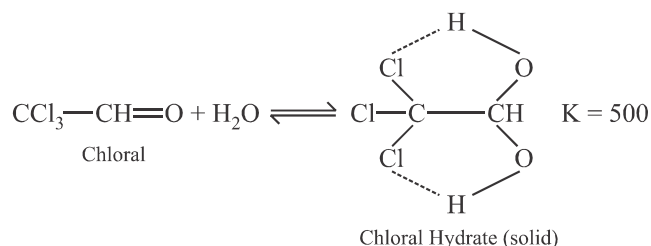
4.1.2 Addition of H_2O – Hydrate Formation

Hydrate is a geminal diol, which is usually unstable, that's why it cannot be isolated from the aqueous solution. Formation depends on the equilibrium constant, which reflects the stability of the corresponding hydrate.



Electron withdrawing groups stabilize the hydrate.

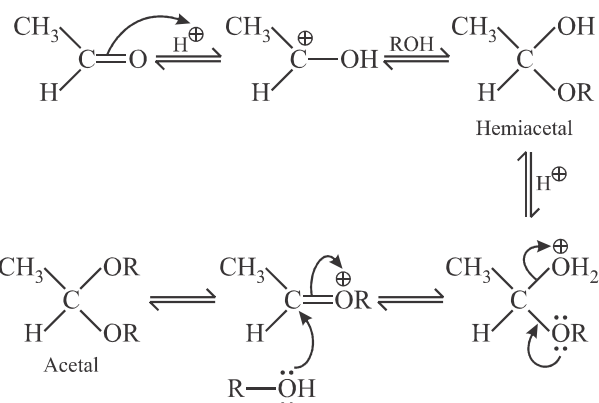
Example - 32



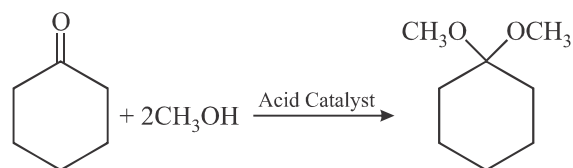
4.1.3 Addition of ROH – Hemiacetal/Acetal Formation

Addition of one alcohol gives hemiacetal and two alcohol molecules gives acetal. Acetal formation involves - acid-catalysed nucleophilic addition to the carbonyl and following unimolecular substitution ($\text{S}_{\text{N}}1$) – protonation, loss of OH^- and attack of alcohol.

Usually reactions are reversible. Simple aldehydes favour the formation of acetal but the hindered aldehydes and most of the ketones do not favour such a formation.

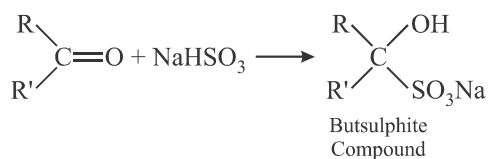


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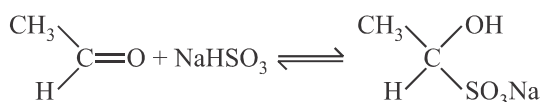


4.1.4 Addition of NaHSO_3 – Bisulphite formation

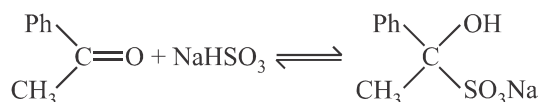
Aldehydes and ketones add on sodium hydrogen sulphite to form a bisulphite compound (hydroxy sulphonate salts). Most of ketones, especially hindered ketones have very less tendency to form bisulphites. Bisulphite compounds are usually crystalline solids insoluble in sodium hydrogen sulphite solution.



Example - 34



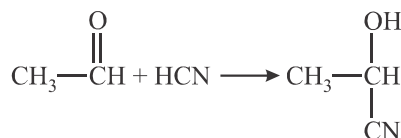
Example - 35



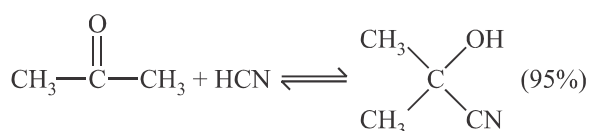
4.1.5 Addition of HCN – Cyanohydrin formation

Addition of HCN to aldehyde or ketone gives cyanohydrin (α -hydroxynitrile). HCN (Hydrocyanic acid) is a weak $\text{Nu}^\ominus$ provider, reaction is usually acid or base catalysed. Presence of $\text{H}^\oplus$ increases reactivity of carbonyl group while $\text{OH}^\ominus$ increases reactivity of $\text{CN}^\ominus$, but in both the conditions, the reaction is a nucleophilic addition. Reaction is usually reversible, cyanohydrin formation depends on the extent of equilibrium (equilibrium constant).

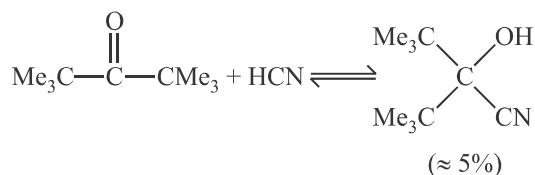
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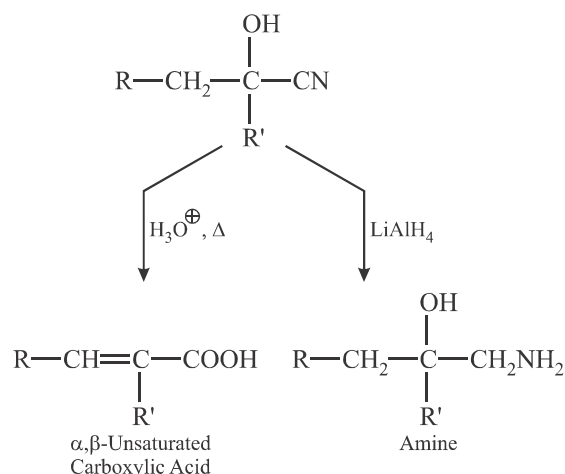
Example - 37



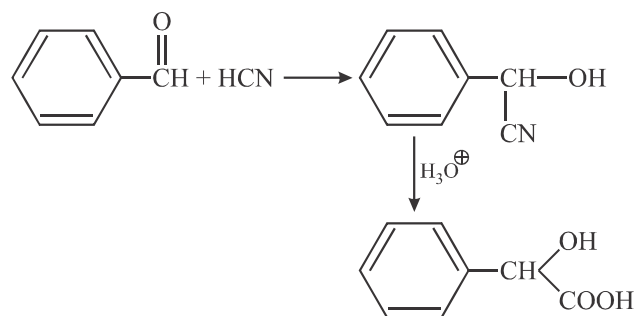
Example - 38



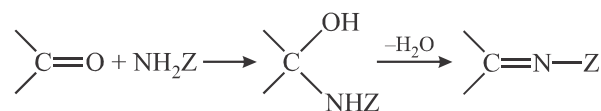
Applications of Cyanohydrins



Example - 39

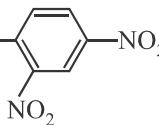
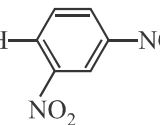


4.1.6 Addition of NH_3 Derivatives



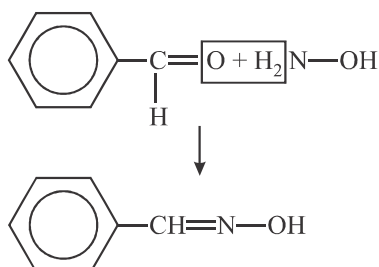
Usually, the reaction is acid catalysed.

The product can demonstrate geometrical isomerism (syn-anti). It is a crystalline solid.

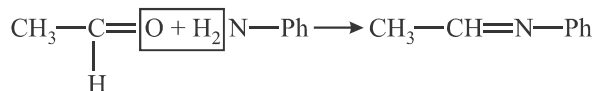
Reagent	Reagent Name	Product	Product Name
$\text{NH}_2\text{—R}$	Primary (1°) Amine	>C=N—R,	Imine (Schiff Base)
$\text{NH}_2\text{—OH}$	Hydroxylamine	>C=N—OH,	Oxime
$\text{NH}_2\text{—NH}_2$	Hydrazine	$\text{>C=N—NH}_2,$	Hydrazone
$\text{NH}_2\text{—NHPh}$	Phenyl Hydrazine	>C=N—NHPh,	Phenyl Hydrazone
$\text{NH}_2\text{—NH—}$ 	2, 4-Dinitrophenyl Hydrazine (Brady's Reagent)	>C=N—NH— 	2, 4-Dinitrophenyl Hydrazone
$\text{NH}_2\text{—NH—C(=O)—NH}_2$	Semicarbazine	$\text{>C=N—NH—C(=O)—NH}_2,$	Semicarbazone

Note... The reaction with Brady's reagent is known as **DNP Test** and is used as a test for detection of carbonyl compounds.

Example - 40



Example - 41

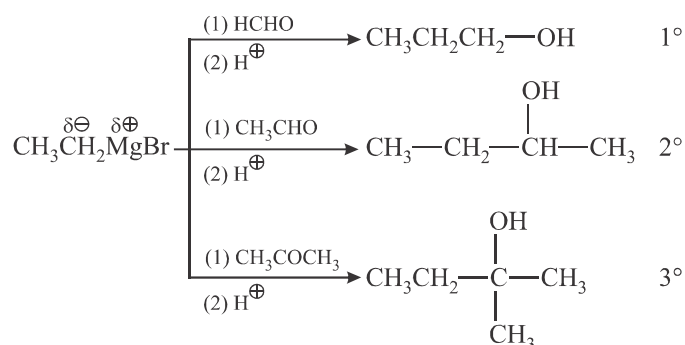


4.1.7 Addition of Grignard Reagent – Alcohol Formation

A Grignard reagent (a strong nucleophile resembling a carbanion $\text{R}^\ominus$) attacks on an aldehyde or a ketone, the subsequent protonation gives the alcohol.

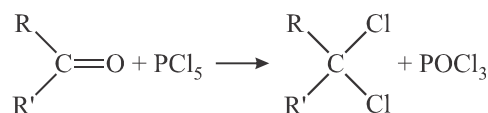


Example - 42



4.1.8 Addition of PCl_5

On reaction with PCl_5 , carbonyl compounds give gem-dihalide.



4.2 Reduction Reactions

4.2.1 Reduction to Alkanes

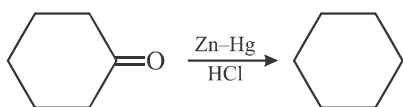
(A) Wolff-Kishner Reduction

Hydrazone or semicarbazone when heated at 180°C with sodium ethoxide, gives out nitrogen and hydrocarbon.

(B) Clemmensen Reduction

Aldehydes and ketones which are not acid sensitive when heated with excess of amalgamated zinc (Zn–Hg) and HCl, give the corresponding alkanes.

Example - 43



4.2.2 Reduction to Alcohols

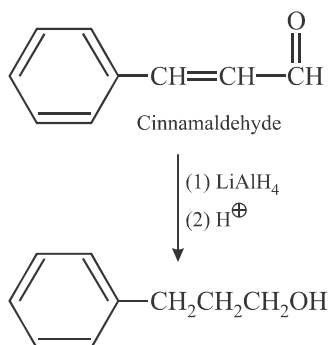
(A) Metal Hydrides and Hydrogenation

Aldehydes and ketones are reduced to 1° and 2° alcohols respectively by various reagents, e.g., H_2/Ni , B_2H_6 , LiAlH_4 , NaBH_4 , etc. Usually LiAlH_4 doesn't affect the double and triple bonds. It selectively reduces carbonyl group but reduces even the double bond in the cinnamyl system.

Example - 44



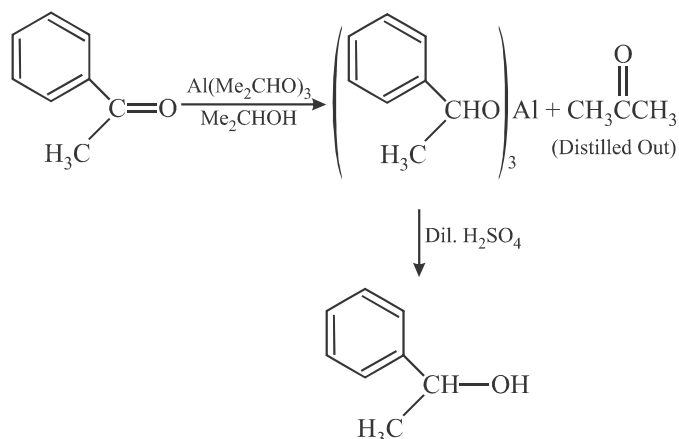
Example - 45



(B) Meerwein-Ponndorf-Verley Reduction

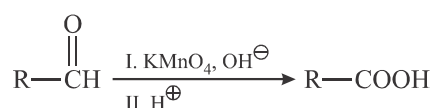
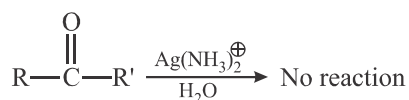
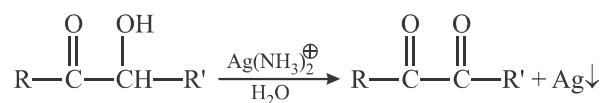
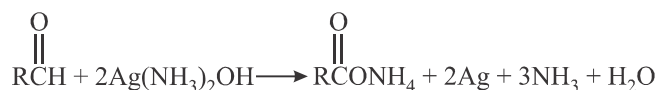
Carbonyl compound is heated with aluminium isopropoxide in isopropanol. Isopropoxide is oxidised to acetone. Without affecting double or triple bonds and some other functional groups, selectively carbonyl group is reduced to the corresponding alcohol.

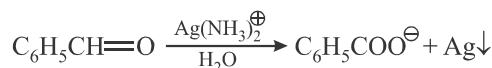
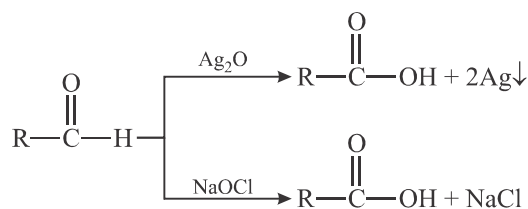
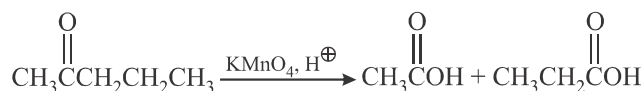
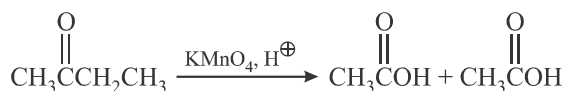
Example - 46



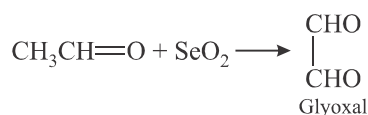
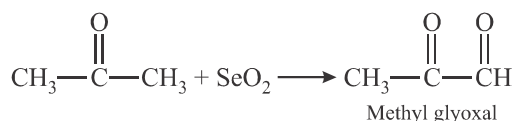
4.3 Oxidation Reactions

Aldehydes are easily oxidised and hence, are powerful reducing agents. Aldehydes reduce Fehling's solution (alkaline solution of Cu^{2+} with tartaric acid) to red cuprous oxide, Tollen's reagent (ammoniacal silver nitrate) to metallic silver. Benzaldehyde do not reduce Cu^{2+} salt (Fehling solution). Ketones (except α -hydroxy ketones) don't reduce Tollen's reagent and the Fehling solution. Benzaldehyde gives positive Tollen's test. All aldehydes and ketones are oxidised by permanganate, acid dichromates etc. These strong oxidising agents oxidise adjacent carbon atom of ketonic carbonyl and the carbonyl group remains chiefly with smaller alkyl group. (**Popoff's rule**)

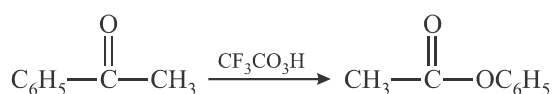
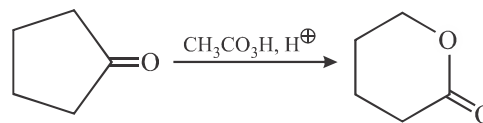
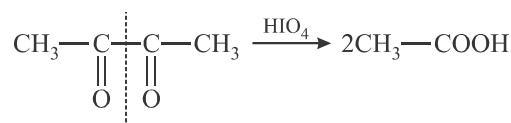


Example - 47

Example - 48

4.3.1 Oxidation by SeO_2

Methyl or methylene adjacent to carbonyl group is oxidised to carbonyl group by SeO_2 .

Example - 49

Example - 50

4.3.2 Baeyer-Villiger Oxidation of Ketones

Ketones on treatment with peracids such as perbenzoic acid or peracetic acid or with other peroxy compounds in the presence of acid catalysts, give carboxylic esters by insertion of oxygen. The reaction involves rearrangement at electron deficient oxygen. For unsymmetrical ketones the approximate order of migratory aptitude is : 3° alkyl $>$ 2° alkyl, phenyl $>$ 1° alkyl.

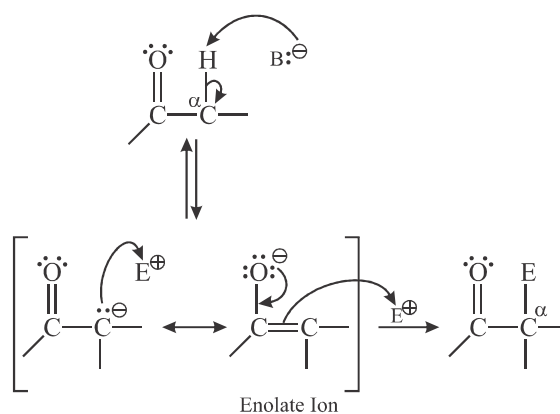
Example - 51

Example - 52

4.3.3 Oxidation by $\text{HIO}_4/\text{Pb}(\text{CH}_3\text{COO})_4$
Example - 53

4.4 Alpha Substitutions

Alpha (α) substitutions involve the replacement of a hydrogen atom at the α carbon atom (the carbon next to the carbonyl group) by some other group. The α hydrogen is more acidic because the enolate ion that results from its removal is resonance-stabilized, with the negative charge delocalized over the α carbon atom and the carbonyl oxygen atom.

Base Catalyzed α -Substitution

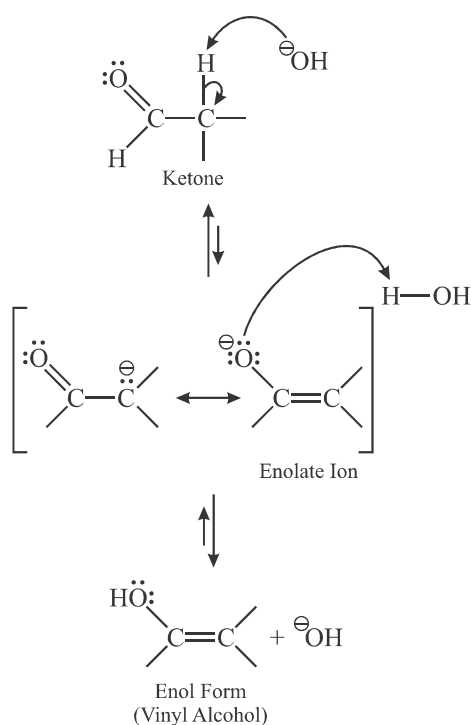
Step 1: Deprotonation of α carbon to form an enolate.

Step 2: Nucleophilic attack on an electrophile.


4.4.1 Keto-Enol Tautomerism
(A) Base-Catalyzed Mechanism

Step 1: Deprotonation of α C.

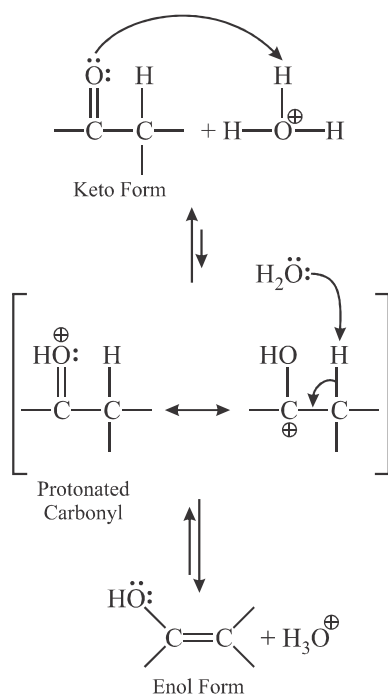
Step 2: Reprotonation on O.



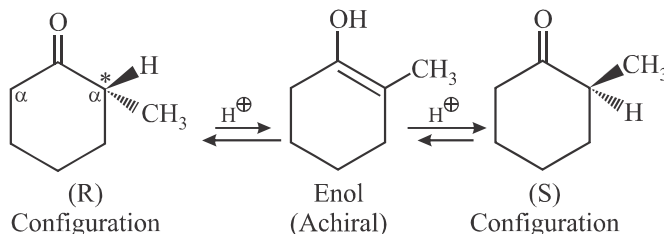
(B) Acid-Catalyzed Mechanism

Step 1: An acid protonates the carbonyl oxygen.

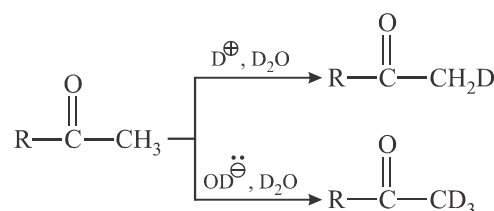
Step 2: Deprotonation on C gives the enol form.



- Note...*
1. Acid catalyzed mechanism is slower than the base catalyzed mechanism.
 2. If an asymmetric carbon atom has an enolizable hydrogen atom, a trace of acid or base allows that carbon to invert its configuration, with the enol serving as the intermediate. A racemic mixture (or an equilibrium mixture of diastereomers) is the result.



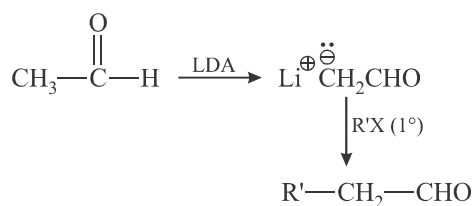
4.4.2 α -Deuteration



- Note...*
1. In acid-catalyzed alpha substitution, only mono substitution is observed.
 2. In base-catalyzed alpha substitution, all α -H are substituted.

4.4.3 α -Alkylation

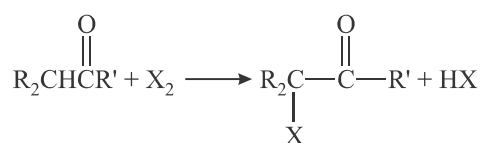
Example - 54



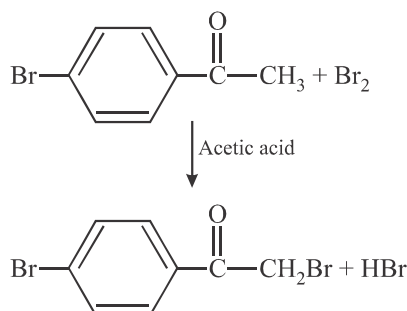
LDA - Lithium Diisopropyl amide $\text{LiN}(\text{iPr})_2$

4.4.4 α -Halogenation

Chlorine or bromine replaces one or more α -hydrogen atoms in aldehydes and ketones. Acetone may be brominated in glacial acetic acid to give monobromoacetone. Reaction occurs by electrophilic attack of the halogen on the carbon-carbon double bond of the enol form. The halogenation is catalysed by acids and bases.



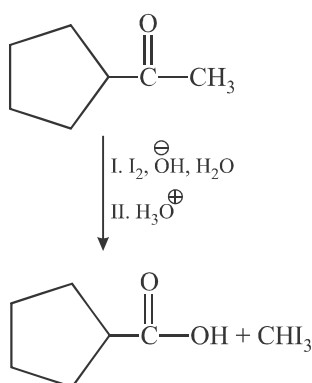
Example - 55



4.4.5 Haloform Reaction

Methyl ketones are cleaved on reaction with excess halogen in the presence of a base. The products are a trihalomethane (haloform) and a carboxylate salt. In aqueous sodium hydroxide, **the rate has been shown to be independent of the bromine concentration**. The reaction is used for structural determinations. The positive iodoform (bright yellow ppt) confirms $\text{CH}_3\text{C}(=\text{O})-$ part usually.

Example - 56

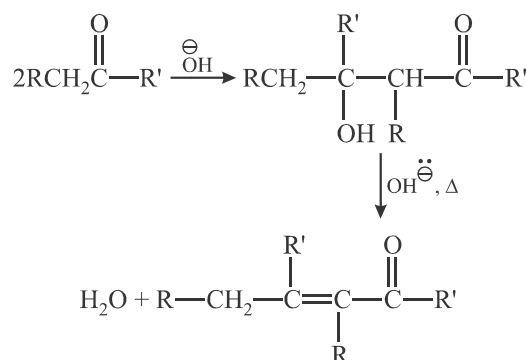


4.5 Special Reactions

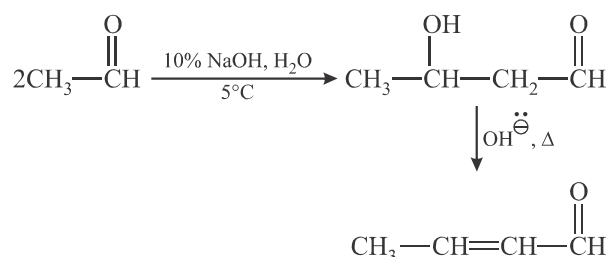
4.5.1 Aldol Condensation

(A) Self-Aldol Condensation

The enolisable carbonyl compound in the presence of H^+ or OH^- (at or below room temp.) undergoes dimerisation, producing aldol. The following dehydration gives α,β -unsaturated aldehyde or ketone. Actually reaction involves nucleophilic addition of an enolate ion to the carbonyl group. Aldehydes are more suitable for aldol condensation than ketones.



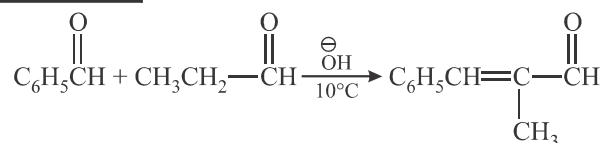
Example - 57



(B) Cross-Aldol Condensation (Claisen-Schmidt Reaction)

A mixed aldol condensation, more precisely the condensation of aromatic aldehyde (or aldehyde without α -Hs) with an enolisable aldehyde and ketone, which is most suitable.

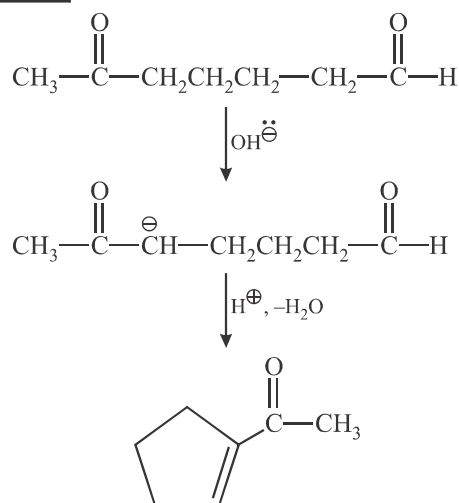
Example - 58



(C) Intramolecular Aldol Condensation

Dicarbonyl compounds give this reaction and undergo cyclization. The formation of the product depends upon the size of the ring (usually 5 or 6 membered rings are formed).

Example - 59

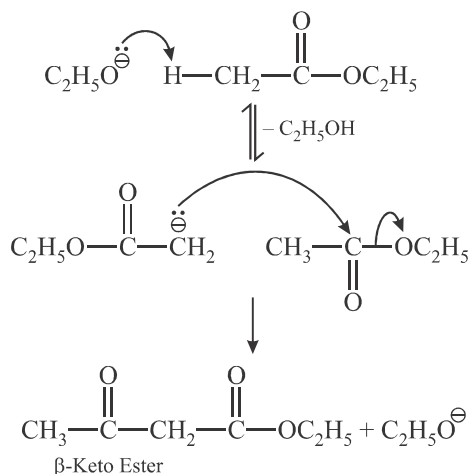


4.5.2 Claisen Condensation

Esters having α -hydrogens on treatment with a strong base such as $\text{C}_2\text{H}_5\text{ONa}$ undergo condensation reactions to produce β -keto esters.

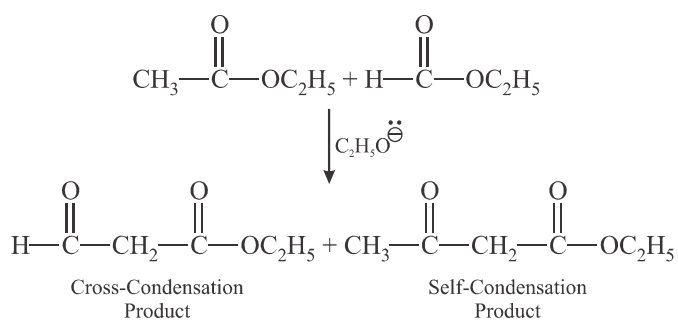
(A) Self-Claisen Condensation

Example - 60



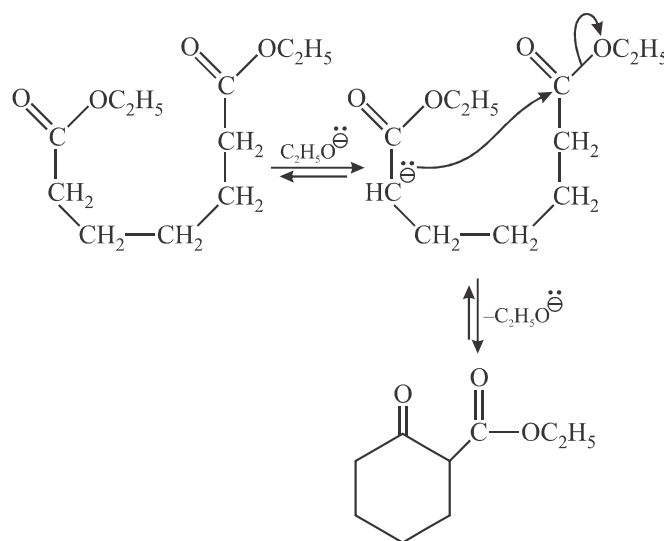
(B) Cross-Claisen Condensation

Example - 61



(C) Intramolecular-Claisen Condensation

(Dieckmann Condensation)

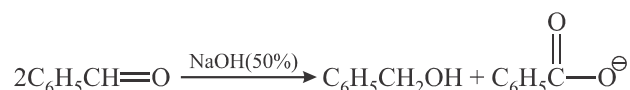


4.5.3 Cannizzaro Reaction

It is a characteristic of aldehydes having no α hydrogens e.g., aromatic aldehydes, formaldehydes, etc. It involves oxidation and reduction of molecules in the presence of highly concentrated OH^- (50%). Reaction is intermolecular. The hydride transfer is the rate determining step.

(A) Self-Cannizzaro Reaction

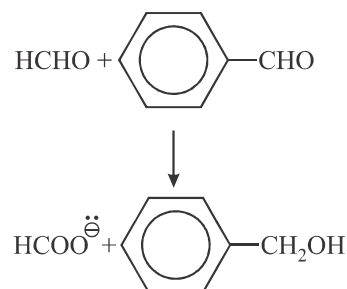
Example - 62



(B) Cross-Cannizzaro Reaction

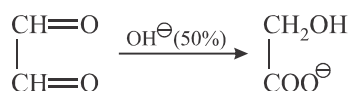
In cross-cannizzaro reaction, acid is formed from that molecule which has more electron deficient carbon.

Example - 63



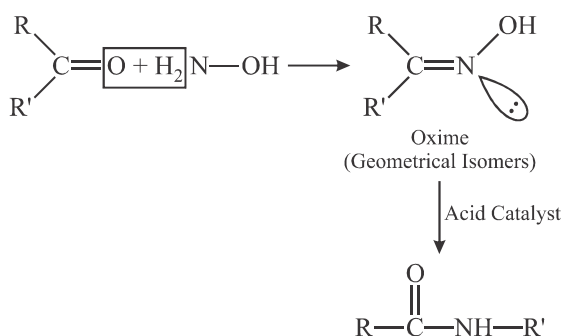
(C) Intramolecular Cannizzaro Reaction

Example - 64



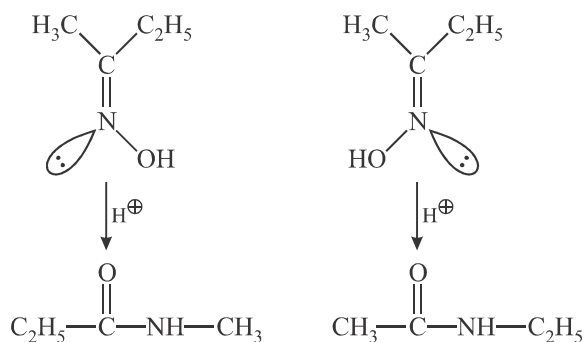
4.5.4 Beckmann Rearrangement

It is the rearrangement of oximes to form substituted amides in the presence of acid catalyst such as H_2SO_4 , P_2O_5 , BF_3 , etc.



Note... The alkyl group that migrates must be **anti** to the OH group.

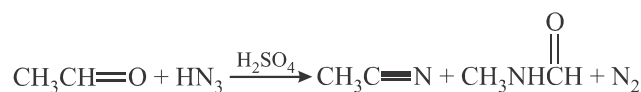
Example - 65



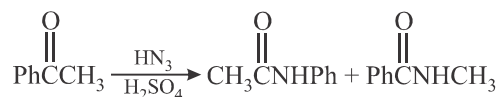
4.5.5 Schmidt Reaction

The reaction involves treatment of hydrazoic acid (HN_3) with carbonyl compounds, in the presence of concentrated sulphuric acid. It is a rearrangement at electron deficient nitrogen and is intramolecular in nature.

Example - 66



Example - 67

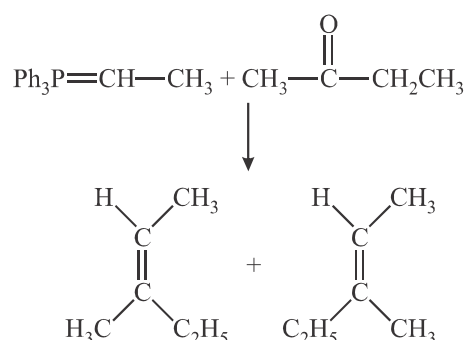


4.5.6 Wittig Reaction

Carbonyl compounds react with phosphorus ylides (phosphoranes) to yield alkenes and triphenyl phosphine oxide. Reaction involves : First, $\text{S}_{\text{N}}2$ attack of triphenyl phosphine (PPh_3) over 1° and 2° alkyl halide – alkyltriphenyl phosphonium salt. Second, alkyl lithium or phenyl lithium removes a proton (acid-base reaction) from salt to give ylide. Third, the combination of ylide with carbonyl group.

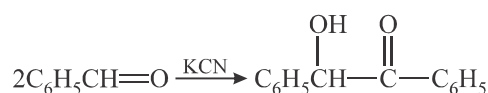
Mixture of E and Z forms are observed.

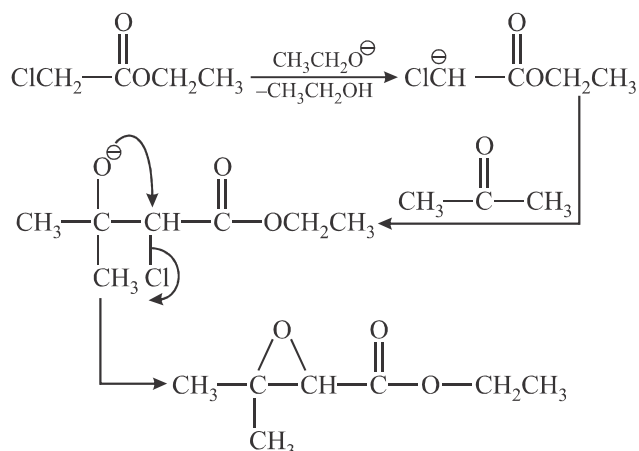
Example - 68



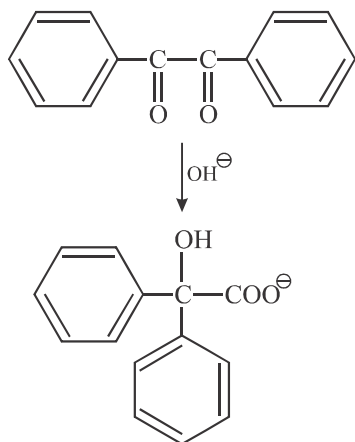
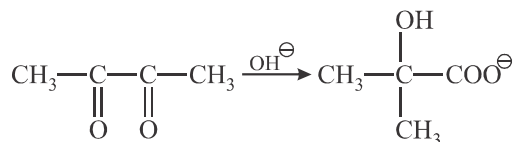
4.5.7 Benzoin Condensation

Benzaldehyde on treatment with aqueous ethanolic potassium cyanide gives benzoin. Mixture of aldehydes give mixed benzoin. Aliphatic aldehydes do not undergo the benzoin condensation. Benzoin is an α -hydroxyketone, hence it reduces Fehling's solution.



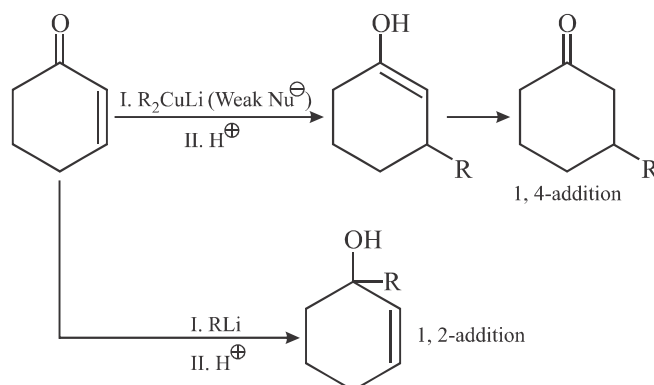
Example - 73

4.5.12 Benzil-Benzilic Acid Rearrangement

α -diketones on treatment with base rearrange to give the salts of α -hydroxyacids. Most known examples are aromatic but can also be applied to aliphatic diketones and to α -keto aldehydes.

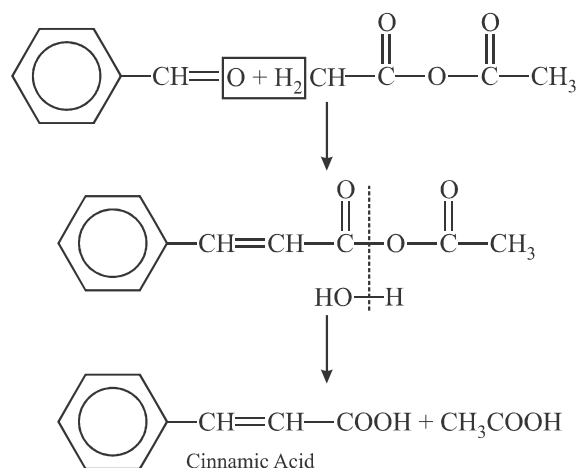
Example - 74

Example - 75

4.5.13 Michael Addition

The strong Nu[⊖] (nucleophile) gives 1, 2-addition (across carbonyl group) and weaker nucleophile gives 1, 4-addition. Active methylene carbanion (Z—CH₂—Z, i.e., Z = electron withdrawing

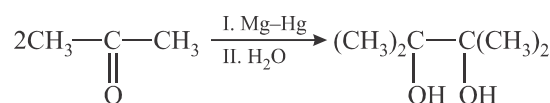
group) is considered a weak nucleophile. The conjugate addition of enolate anions to α , β -unsaturated carbonyl compounds is known as **Michael Additions**.

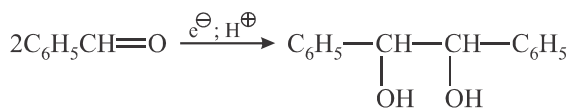
Example - 76

4.5.14 Perkin Addition

It is the condensation of an aromatic aldehyde with an acid anhydride to form α , β -unsaturated carboxylic acids.

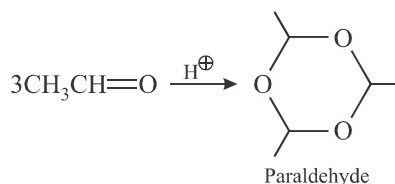
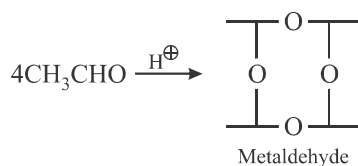
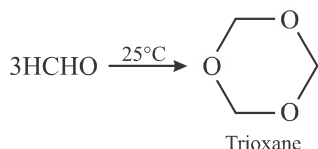
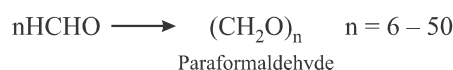
Example - 77

4.5.15 Reduction with Metals

The reduction of ketones with magnesium amalgam gives pinacol. Similarly, benzaldehyde on treatment with zinc and hydrochloric acid or with sodium amalgam gives hydrobenzoin.

Example - 78


Example - 79**4.5.16 Polymerisation**

Formaldehyde solution is evaporated to dryness – a solid polymeric formaldehyde in the form of polymethylene glycol appears, called paraformaldehyde. In the gaseous stage, at room temperature a white crystalline solid metaformaldehyde forms. Acetaldehyde forms metaldehyde and paraldehyde.

**5. ANALYSIS OF ALDEHYDES AND KETONES**

Aldehydes and ketones are characterized through the addition to the carbonyl group of nucleophilic reagents, especially derivatives of ammonia. An aldehyde or ketone will, for example, react with 2, 4-dinitrophenylhydrazine to form an insoluble yellow or red solid.

Aldehydes are characterized, and in particular are differentiated from ketones, through their ease of oxidation : aldehydes give a positive test with Tollens' reagent ; ketones do not. A positive Tollen's test is also given by a few other kinds of easily oxidized compounds, e.g., certain phenols and amines; these compounds do not, however, give positive tests with 2, 4-dinitrophenylhydrazine.

Aldehydes are also, of course, oxidized by many other oxidizing agents : by cold, dilute, neutral KMnO_4 and by CrO_3 in H_2SO_4 .

A highly sensitive test for aldehydes is the Schiff test. An aldehyde reacts with the fuchsin-aldehyde reagent to form a characteristic magenta color.

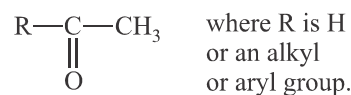
Aliphatic aldehydes and ketones having an α -hydrogen react with Br_2 in CCl_4 . This reaction is generally too slow to be confused with a test for unsaturation, and moreover it liberates HBr .

Aldehydes and ketones are generally identified through the melting points of derivatives like 2, 4-dinitrophenylhydrazones, oximes, and semicarbazones.

Methyl ketones are characterized through the iodoform test.

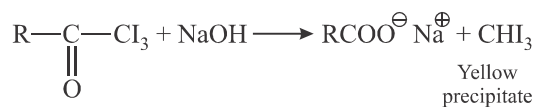
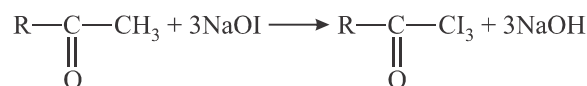
5.1 Iodoform Test

Whether or not a ketone is a methyl ketone is shown by the iodoform test. The ketone is treated with iodine and sodium hydroxide (sodium hypoiodite, NaOI); a ketone of the structure.

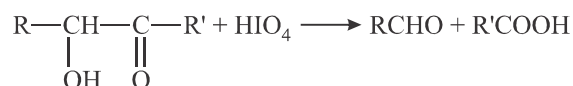
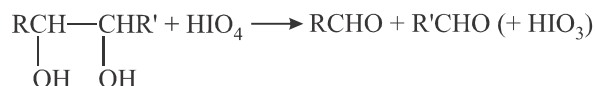
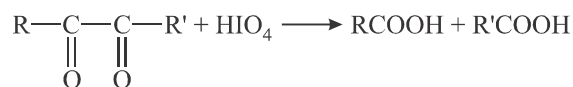


yields a yellow precipitate of iodoform (CHI_3 , m.p. 119°C).

The reaction involves halogenation and cleavage :

**5.2 Periodic Acid Oxidation**

Upon treatment with periodic acid, HIO_4 , compounds containing two or more $=\text{O}$ or $-\text{OH}$ groups attached to adjacent carbon atoms undergo oxidation with cleavage of carbon-carbon bonds. For example :



The oxidation is particularly useful in determination of structure, as we shall find in our study of carbohydrates. Qualitatively, oxidation by HIO_4 is indicated by formation of a white precipitate (AgIO_3) upon addition of silver nitrate. Since the reaction is usually

quantitative, valuable information is given by the nature and amounts of the products, and by the quantity of periodic acid consumed.

6. IMPORTANT CARBONYL COMPOUNDS

6.1 FORMALDEHYDE – HCHO

It is present in green leaves of plants where its presence is supposed to be due to the reaction of CO_2 with water in presence of sunlight and chlorophyll.



Traces of formaldehyde are formed when incomplete combustion of wood, sugar, coal, etc., occurs.

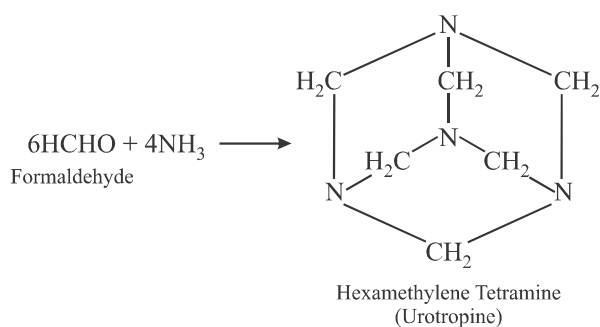
Formaldehyde is prepared in laboratory by catalytic oxidation of methyl alcohol.

Excess of methyl alcohol is removed by fractional distillation. The resulting mixture is called formalin which contains 40% formaldehyde, 8% methyl alcohol and 52% water.

6.1.1 Important Reactions

Reaction with ammonia

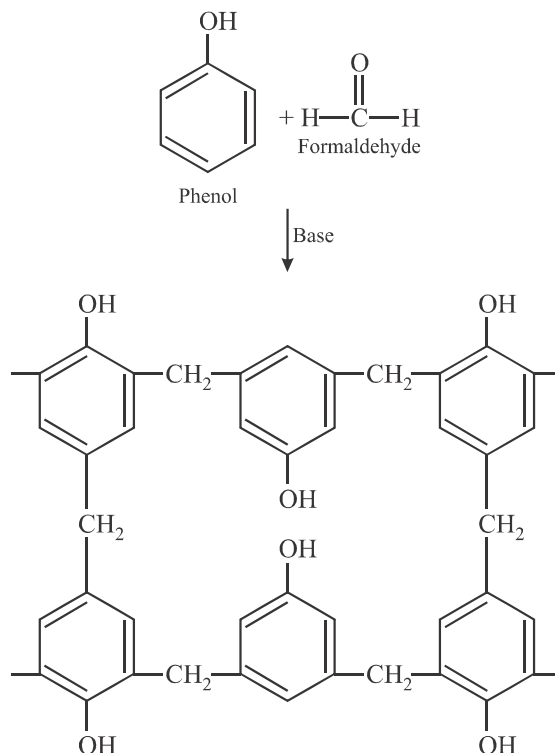
Like other aldehydes, formaldehyde does not form addition product with ammonia but crystalline compound, hexamethylene tetramine (Urotropine).



It is used as medicine in case of urinary troubles under the name of **Urotropine** or hexamine. Nitration of urotropine under controlled conditions gives an explosive **RDX (Research and Development Explosive)**.

Condensation with phenol

Formaldehyde condenses with phenol to give a synthetic plastic, bakelite. The condensation occurs in presence of dilute sodium hydroxide or ammonia at $80^\circ\text{--}90^\circ\text{C}$. Bakelite is used for preparing electrical insulators, electric switches and toys, etc.



6.1.2 Uses

- The 40% solution of formaldehyde in water (formalin) is used as disinfectant, germicide and antiseptic. It is used for the preservation of biological specimens.
- It is used in the preparation of hexamethylene tetramine (urotropine) which is used as an antiseptic and germicide.
- It is used in silvering of mirror and decolouring vat dyes.
- It is employed in the manufacture of synthetic dyes such as para-rosaniline, indigo, etc.
- It is used in the manufacture of **formamint** (by mixing formaldehyde with lactose) – a throat lozenge.

6.2 ACETALDEHYDE – CH_3CHO

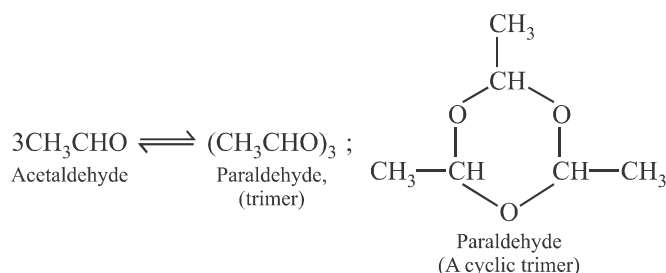
6.2.1 Important Reactions

Polymerisation

Acetaldehyde undergoes polymerisation forming different products under different conditions.

Paraldehyde

It is formed, when anhydrous acetaldehyde is treated with a few drops of conc. sulphuric acid.



It is a pleasant smelling liquid (B.P. 124°C). It has cyclic structure and when heated with dilute sulphuric acid, it changes again into acetaldehyde. It is used in the medicine as a hypnotic and soporific (sleep producing).

6.2.2 Uses

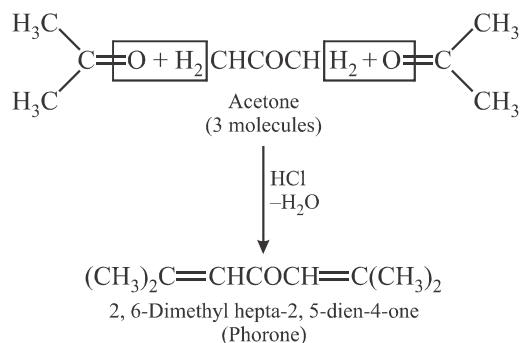
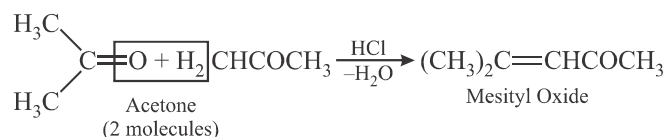
Acetaldehyde is used :

- In the preparation of acetic acid, acetic anhydride, ethyl acetate, chloral, buta-1,3-diene (used in rubbers), dyes and drugs.
- As an antiseptic inhalant in nose troubles.
- In the preparation of paraldehyde (hypnotic) and metaldehyde (solid fuel).
- In the preparation of acetaldehyde ammonia (a rubber accelerator).
- As a reducing agent in silvering of mirror.

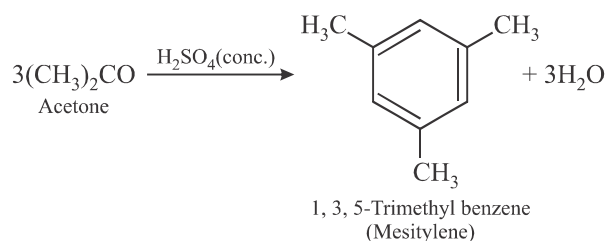
6.3 ACETONE – CH_3COCH_3

6.3.1 Important Reactions

When acetone saturated with HCl gas is kept at low temperature for few days, it forms mesityl oxide and phorone which can be separated by distillation.



When acetone is distilled with concentrated sulphuric acid it forms mesitylene.

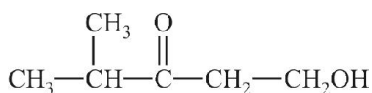


6.3.2 Uses

- As a solvent for cellulose acetate, cellulose nitrate, celluloid, lacquers, resins, etc.
- For storing acetylene.
- In the manufacture of cordite—a smokeless powder explosive.
- In the preparation of chloroform, iodoform, sulphonal and chloretone.
- As a nail polish remover.
- In the preparation of an artificial scent (ionone), plexiglas (a tough transparent plastic) and synthetic rubber.

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS**Nomenclature**

1. The IUPAC name of $\text{Cl}_3\text{CCH}_2\text{CHO}$ is :
(a) chloral
(b) 3,3,3-trichloropropanal
(c) 1,1,1-trichloropropanal
(d) 2,2,2-trichloropropanal
2. The IUPAC name of $\text{CH}_3\text{COCH}(\text{CH}_3)_2$ is :
(a) isopropyl methyl ketone
(b) 2-methyl-3-butanone
(c) 4-methyl-isopropyl ketone
(d) 3-methyl-2-butanone
3. The IUPAC name of the following structure is



- (a) 1-hydroxy 4-methyl 3-pentanone
(b) 2-methyl 5-hydroxy 3-pentanone
(c) 4-methyl 3-oxo 1-pentanol
(d) Hexanol-1, one-3

General Characteristics

4. The general formula of both aldehydes and ketones is :
(a) $\text{C}_n\text{H}_{2n+2}\text{O}$ (b) $\text{C}_n\text{H}_{2n}\text{O}$
(c) $\text{C}_n\text{H}_{2n-2}\text{O}$ (d) $\text{C}_n\text{H}_{2n+4}\text{O}$
5. Aldehydic group can occur :
(a) any where in the carbon chain
(b) in the middle of carbon chain
(c) only at the second carbon atom of the chain
(d) only at the end carbon atom of the carbon chain
6. Formalin is an aqueous solution of :
(a) formic acid (b) fluorescein
(c) formaldehyde (d) furfuraldehyde

7. Nucleophilic addition reaction will be most favoured in :

- (a) $\text{CH}_3\text{CH}_2\text{CHO}$
(b) CH_3CHO
(c) $\text{CH}_3\text{CH}_2\text{CH}_2 - \overset{\text{O}}{\underset{||}{\text{C}}} - \text{CH}_3$
(d) $(\text{CH}_3)_2\text{C} = \text{O}$

8. Reaction of acetaldehyde with HCN followed by hydrolysis gives a compound which shows

- (a) Optical isomerism (b) Geometrical isomerism
(c) Metamerism (d) Tautomerism

9. Ethanedial has which functional group(s)

- (a) One ketonic (b) Two aldehydic
(c) One double bond (d) Two double bond

10. Which of the aldehyde is most reactive ?

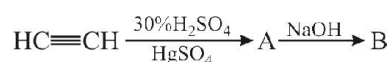
- (a) $\text{C}_6\text{H}_5 - \text{CHO}$ (b) CH_3CHO
(c) HCHO (d) All the equally reactive

11. Correct order of reactivity of CH_3CHO , $\text{C}_2\text{H}_5\text{COCH}_3$ and CH_3COCH_3 is

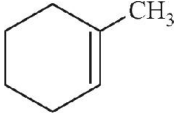
- (a) $\text{CH}_3\text{CHO} > \text{CH}_3\text{COCH}_3 > \text{CH}_3\text{COC}_2\text{H}_5$
(b) $\text{C}_2\text{H}_5\text{COCH}_3 > \text{CH}_3\text{COCH}_3 > \text{CH}_3\text{CHO}$
(c) $\text{CH}_3\text{COCH}_3 > \text{CH}_3\text{CHO} > \text{C}_2\text{H}_5\text{COCH}_3$
(d) $\text{CH}_3\text{COCH}_3 > \text{C}_2\text{H}_5\text{COCH}_3 > \text{CH}_3\text{CHO}$

Preparation

12. Predict the product 'B' in the sequence of reaction



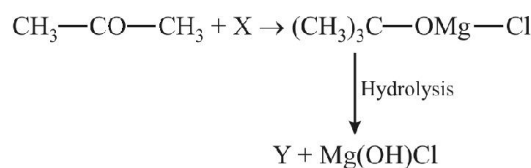
- (a) CH_3COONa (b) CH_3COOH
(c) CH_3CHO (d) $\text{CH}_3 - \underset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_2\text{CHO}$

13.  On reductive ozonolysis yields
- (a) 6-oxoheptanal (b) 6-oxoheptanoic acid
(c) 6-hydroxyheptanal (d) 3-hydroxypentanal
14. Benzaldehyde + NaOH →
- (a) Benzyl alcohol (b) Benzoic alcohol
(c) Hydrobenzamide (d) Cinnamic acid
15. When but-2-yne is treated dil. $\text{H}_2\text{SO}_4/\text{HgSO}_4$ the product formed is :
- (a) butan-1-ol (b) butan-2-ol
(c) acetone (d) butanone
16. The end product in the following sequence of reaction is
- $$\text{HC} \equiv \text{CH} \xrightarrow[20\% \text{ H}_2\text{SO}_4]{1\% \text{ HgSO}_4} \text{A} \xrightarrow{\text{CH}_3\text{MgX}} \text{B} \xrightarrow{[\text{O}]}$$
- (a) Acetic acid (b) Isopropyl alcohol
(c) Acetone (d) Ethanol
17. Which of the following compound gives a ketone with Grignard reagent
- (a) Formaldehyde (b) Ethyl alcohol
(c) Methyl cyanide (d) Methyl iodide

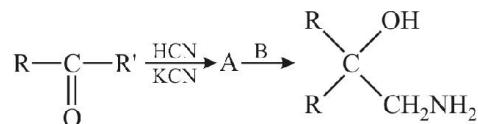
Chemical Reactions

18. Dry heating of calcium acetate gives
- (a) Acetaldehyde (b) Ethane
(c) Acetic acid (d) Acetone
19. On heating calcium acetate and calcium formate, the product formed is
- (a) CH_3COCH_3 (b) CH_3CHO
(c) $\text{HCHO} + \text{CaCO}_3$ (d) $\text{CH}_3\text{CHO} + \text{CaCO}_3$

20. Identify the reactant X and the product Y



- (a) $\text{X} = \text{MgCl}_2$; $\text{Y} = \text{CH}_3\text{CH} = \text{CH}_2$
(b) $\text{X} = \text{CH}_3\text{MgCl}$; $\text{Y} = \text{C}_2\text{H}_5\text{COCH}_3$
(c) $\text{X} = \text{CH}_3\text{MgCl}$; $\text{Y} = (\text{CH}_3)_3\text{C—OH}$
(d) $\text{X} = \text{C}_2\text{H}_5\text{MgCl}$; $\text{Y} = (\text{CH}_3)_3\text{C—OH}$
21. A and B in the following reactions are



- (a) $\text{A} = \begin{array}{c} \text{R} \quad \text{CN} \\ \diagdown \quad \diagup \\ \text{C} \\ \diagup \quad \diagdown \\ \text{R}' \quad \text{OH} \end{array}$, $\text{B} = \text{LiAlH}_4$
(b) $\text{A} = \begin{array}{c} \text{R} \quad \text{OH} \\ \diagdown \quad \diagup \\ \text{C} \\ \diagup \quad \diagdown \\ \text{R}' \quad \text{COOH} \end{array}$, $\text{B} = \text{NH}_3$
(c) $\text{A} = \begin{array}{c} \text{R} \quad \text{CN} \\ \diagdown \quad \diagup \\ \text{C} \\ \diagup \quad \diagdown \\ \text{R}' \quad \text{OH} \end{array}$, $\text{B} = \text{H}_3\text{O}^+$
(d) $\text{A} = \text{RR}'\text{CH}_2\text{CN}$, $\text{B} = \text{NaOH}$

22. For $\text{C}_6\text{H}_5\text{CHO}$ which of the following is incorrect
- (a) On oxidation it yields benzoic acid
(b) It is used in perfumery
(c) It is an aromatic aldehyde
(d) On reduction yields phenol
23. Grignard reagent on reaction with acetone forms
- (a) Tertiary alcohol (b) Secondary alcohol
(c) Acetic acid (d) Acetaldehyde

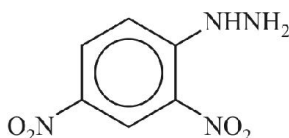
24. Consider the following statement Acetophenone can be prepared by

- (1) Oxidation of 1-phenylethanol
 - (2) Reaction of 1-phenylethanol with methyl magnesium bromide
 - (3) Friedel Craft's reaction of benzene with acetyl chloride
 - (4) Distillation of calcium benzoate
- (a) 1 and 2 (b) 1 and 4
(c) 1 and 3 (d) 3 and 4

25. Which one of the following pairs is not correctly matched

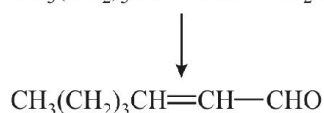
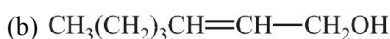
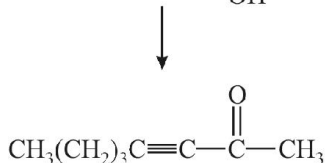
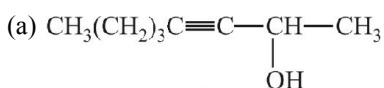
- (a) $>C=O \xrightarrow{\text{Clemenson's reduction}} >CH_2$
(b) $>C=O \xrightarrow{\text{Wolf-Kishner reduction}} >CHOH$
(c) $-COCl \xrightarrow{\text{Rosenmund's reduction}} CHO$
(d) $-C \equiv N \xrightarrow{\text{Stephen reduction}} CHO$

26. Which of the following compounds containing carbonyl group will give coloured crystalline compound with



- (a) CH_3COCl (b) CH_3COCH_3
(c) $CH_3CO(OC_2H_5)$ (d) CH_3CONH_2

27. Which of the following oxidation reactions can be carried out with chromic acid in aqueous acetone at $5-10^\circ C$.



- (c) $CH_3(CH_2)_3CH_2OH \longrightarrow CH_3(CH_2)_3CHO$
(d) None of these

Oxidation Reactions

28. Ketones ($R-\overset{\substack{|| \\ O}}{C}-R_1$) where $R = R_1 =$ alkyl group. It can

be obtained in one step by

- (a) Hydrolysis of esters
(b) Oxidation of primary alcohol
(c) Oxidation of secondary alcohol
(d) Reaction of acid halide with alcohols

29. The oxidation of toluene to benzaldehyde by chromyl chloride is called

- (a) Cannizzaro reaction (b) Wurtz reaction
(c) Etard reaction (d) Reimer-Tiemann reaction

30. Methyl ethyl ketone is prepared by the oxidation of

- (a) 2-propanol (b) 1-butanol
(c) 2-butanol (d) t-butyl alcohol

31. The oxidation product of 2-propanol with hot conc. HNO_3 is

- (a) Ethanoic acid (b) Propanone
(c) Propanal (d) None of these

32. Fehling's test is positive for

- (a) Acetaldehyde (b) Benzaldehyde
(c) Ether (d) Alcohol

33. The reaction of acetaldehyde with conc. $KMnO_4$ gives

- (a) CH_3COOH (b) CH_3CH_2OH
(c) $HCHO$ (d) CH_3OH

34. An aldehyde on oxidation gives

- (a) An alcohol (b) An acid
(c) A ketone (d) An ether

35. Aldehydes are first oxidation products of :

- (a) primary alcohols (b) secondary alcohols
(c) tertiary alcohols (d) monohydric alcohols

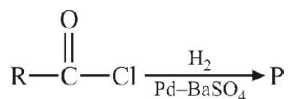
36. Isopropyl alcohol on oxidation forms :

- (a) acetone (b) ether
(c) acetaldehyde (d) ethylene

37. Oxidation of acetaldehyde with selenium dioxide produces:
- (a) ethanoic acid (b) methanoic acid
(c) glyoxal (d) oxalic acid
44. Which of the following reagents converts both acetaldehyde and acetone to alkanes ?
- (a) Ni/H₂ (b) LiAlH₄
(c) I₂/NaOH (d) Zn-Hg/conc.HCl

Reduction Reactions

38. In the following reaction, product P is



- (a) RCH₂OH (b) RCOOH
(c) RCHO (d) RCH₃
39. $\text{CH}_3\text{COCl} \xrightarrow[\text{Pd/BaSO}_4]{2\text{H}} \text{CH}_3\text{CHO} + \text{HCl}$.
- The above reaction is called
- (a) Reimer-Tiemann reaction
(b) Cannizzaro reaction
(c) Rosenmund reaction
(d) Reformatsky reaction
40. Catalyst used in Rosenmund reduction is
- (a) Pd/BaSO₄ (b) Zn-Hg couple
(c) LiAlH₄ (d) Ni/H₂
41. Wolff-Kishner reduction is :
- (a) reduction of carbonyl compounds into hydrocarbons
(b) reduction of carbonyl compounds into alcohols
(c) reduction of nitrobenzene into aniline
(d) reduction of carbohydrates into alcohols
42. Reduction of acetonitrile in presence of SnCl₂/HCl, followed by hydrolysis gives :
- (a) HCHO (b) CH₃CHO
(c) CH₃CH₂CHO (d) CH₃COCH₃
43. If pentan-2-one is reacted with NaBH₄, followed by hydrolysis with D₂O, the product will be :
- (a) CH₃CD(OD)CH₂CH₂CH₃
(b) CH₃CD(OH)CH₂CH₂CH₃
(c) CH₃CH(OH)CH₂CH₂CH₃
(d) CH₃CH(OD)CH₂CH₂CH₃
45. What reagent is used in the Rosenmund's reduction ?
- (a) H₂/Pd-BaSO₄
(b) LiAlH₄
(c) NH₂NH₂/KOH/CH₂OH-CH₂OH
(d) Zn-Hg/HCl

Condensation Reactions

46. Aldol condensation will not be observed in
- (a) Chloral (b) Phenyl acetaldehyde
(c) Hexanal (d) Ethanal
47. Which of the following compound will undergo self aldol condensation in the presence of cold dilute alkali
- (a) C₆H₅CHO (b) CH₃CH₂CHO
(c) CH≡C-CHO (d) CH₂=CH-CHO
48. Which of the following will not undergo aldol condensation
- (a) Acetaldehyde (b) Propanaldehyde
(c) Benzaldehyde (d) Trideuteroacetaldehyde
49. Benzaldehyde + NaOH →
- (a) Benzyl alcohol (b) Benzoic alcohol
(c) Hydrobenzamide (d) Cinnamic acid
50. The following reagent converts C₆H₅COCHO to C₆H₅CHOHCOONa
- (a) Aq. NaOH (b) Acidic Na₂S₂O₃
(c) Na₂CrO₄/H₂SO₄ (d) NaNO₂/HCl
51. Treatment of propionaldehyde with dilute NaOH solution gives
- (a) CH₃CH₂COOCH₂CH₂CH₃
(b) CH₃CH₂CHOHCH(CH₃)CHO
(c) CH₃CH₂CHOHCH₂CH₂CHO
(d) CH₃CH₂COCH₂CH₂CHO

52. Aldol condensation is :
- self condensation of aldehydes only having atleast one alpha hydrogen
 - self condensation of ketones only having atleast one alpha hydrogen
 - self condensation of both aldehydes and ketones having atleast one alpha hydrogen
 - self condensation of both aldehydes and ketones having no alpha hydrogen
53. Cannizzaro's reaction is not given by :
- CH_3CHO
 - HCHO
 - benzaldehyde
 - triethyl acetaldehyde
54. $\text{CH}_3 - \overset{\overset{\text{CH}_3}{|}}{\underset{\underset{\text{CH}_3}{|}}{\text{C}}} - \text{CHO}$ does not show aldol condensation because :
- carbon is bounded by 3-heavy methyl groups
 - due to the absence of α -hydrogen
 - it is a neutral molecule
 - due to hindrance created by methyl group for carbanion
58. Which compound does not form iodoform with alkali and iodine
- Acetone
 - Ethanol
 - Diethyl ketone
 - Isopropyl alcohol
59. Which compound does not gives yellow ppt. with iodine and alkali
- 2-hydroxy propane
 - Acetophenone
 - Methyl ketone
 - Acetamide

Chemical Tests

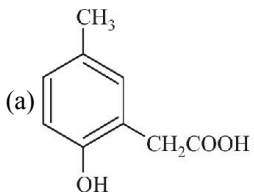
60. Which of the following is incorrect
- FeCl_3 is used in the detection of phenols
 - Fehling solution is used in the detection of glucose
 - Tollen's reagent is used in detection of unsaturation
 - NaHSO_3 is used in the detection of carbonyl compounds
61. $\text{C}_2\text{H}_5\text{CHO}$ and $(\text{CH}_3)_2\text{CO}$ can be distinguished by testing with
- Phenyl hydrazine
 - Hydroxylamine
 - Fehling solution
 - Sodium bisulphite
62. Acetaldehyde cannot show
- Iodoform test
 - Lucas test
 - Benedict's test
 - Tollen's test
63. Silver mirror is a test for
- Aldehydes
 - Thio alcohols
 - Amines
 - Ethers
64. Which of the following does not turn Schiff's reagent to pink
- Formaldehyde
 - Benzaldehyde
 - Acetone
 - Acetaldehyde

Haloform Reaction

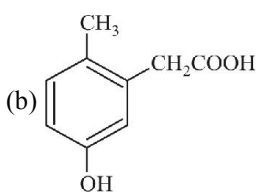
55. Which of the following does not give iodoform test
- $\text{CH}_3\text{CH}_2\text{OH}$
 - CH_3OH
 - CH_3CHO
 - PhCOCH_3
56. Which of the following will not give iodoform test
- Ethanal
 - Ethanol
 - 2-propanone
 - 3-pentanone
57. Which of the following is obtained when chloral is boiled with NaOH
- CH_3Cl
 - CHCl_3
 - CCl_4
 - None of these

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTIONS

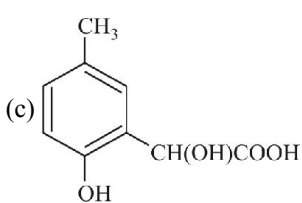
- Which one of the following undergoes reaction with 50% sodium hydroxide solution to give the corresponding alcohol and acid ? (2004)
(a) Phenol (b) Benzaldehyde
(c) Butanal (d) Benzoic acid
- p-cresol reacts with chloroform in alkaline medium to give the compound A which adds hydrogen cyanide to form the compound B. The latter, on acidic hydrolysis gives chiral carboxylic acid. The structure of the carboxylic acid is (2005)



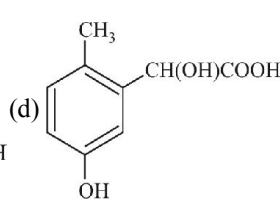
(a)

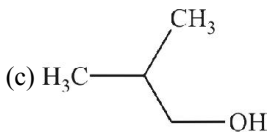


(b)

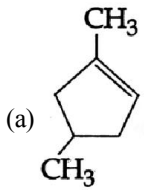


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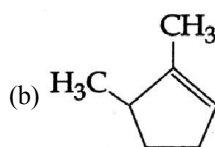


(d)
- Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is (2006)
(a) $CH_3CH_2CH(OH)CH_2CH_3$
(b) $C_6H_5CH_2CH_2OH$
(c) 
(d) $PhCHOHCH_3$
- The increasing order of the rate of HCN addition to compounds A-D is (2006)
(A) HCHO (B) CH_3COCH_3
(C) $PhCOCH_3$ (D) $PhCOPh$
(a) $A < B < C < D$ (b) $D < B < C < A$
(c) $D < C < B < A$ (d) $C < D < B < A$
- Phenol, when it first reacts with concentrated sulphuric acid and then with concentrated nitric acid, gives (2008)
(a) 2, 4, 6-trinitrophenol (b) o-nitrophenol
(c) p-nitrophenol (d) nitrobenzene
- In Cannizaro reaction given below (2009)
$$2PhCHO \xrightarrow{OH^-} PhCH_2OH + PhCO_2^-$$

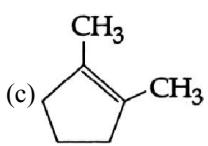
the slowest step is
(a) the attack of OH^- at the carboxyl group
(b) the transfer of hydride to the carbonyl group
(c) the abstraction of proton from the carboxylic group
(d) the deprotonation of $PhCH_2OH$
- Which of the following on heating with aqueous KOH, produces acetaldehyde ? (2009)
(a) CH_3COCl (b) CH_3CH_2Cl
(c) CH_2ClCH_2Cl (d) CH_3CHCl_2
- Silver mirror test is given by which one of the following compounds ? (2011)
(a) Acetaldehyde (b) Acetone
(c) Benzophenone (d) None of these
- Trichloroacetaldehyde was subjected to Cannizaro's reaction by using NaOH. The mixture of the products contains sodium trichloroacetate ion and another compound. The other compound is (2014)
(a) 2, 2, 2-trichloroethanol (b) trichloromethanol
(c) 2, 2, 2-trichloromethanol
(d) chloroform
- Which compound would give 5-keto-2-methyl hexanal upon ozonolysis? (2015)



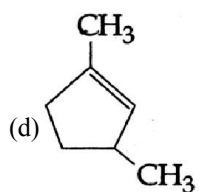
(a)



(b)

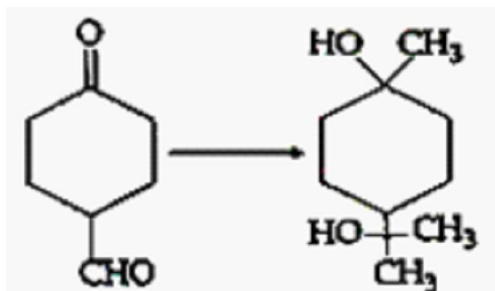


(c)

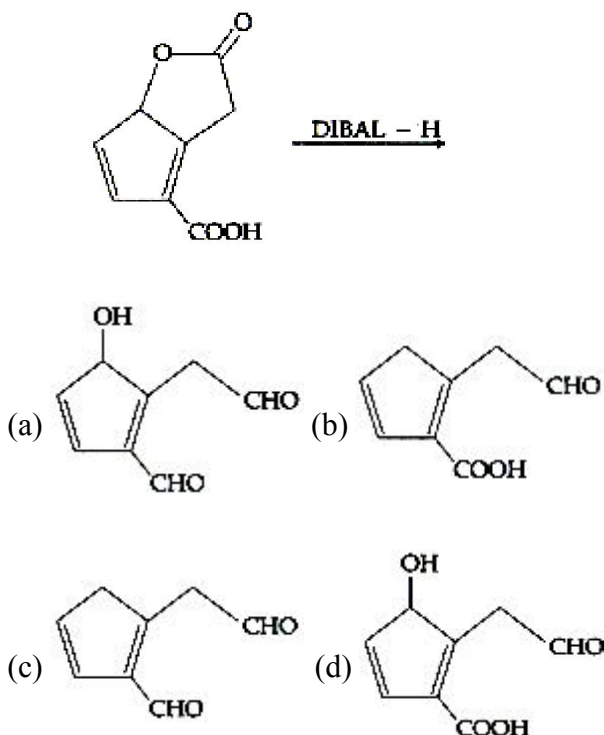


(d)

11. The correct sequence of reagents for the following conversion will be : (2017)



- (a) CH_3MgBr , H^+ / CH_3OH , $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$
 (b) CH_3MgBr , $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$, H^+ / CH_3OH
 (c) $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$, CH_3MgBr , H^+ / CH_3OH
 (d) $[\text{Ag}(\text{NH}_3)_2]^+ \text{OH}^-$, H^+ / CH_3OH , CH_3MgBr
12. The major product obtained in the following reaction is : (2017)



JEE MAINS ONLINE QUESTION

1. Which is the major product formed when acetone is heated with iodine and potassium hydroxide?

Online 2014 SET (1)

- (a) Iodoacetone (b) Acetic acid
 (c) Iodoform (d) Acetophenone

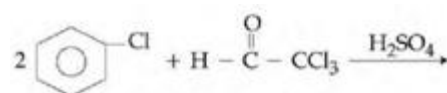
2. Tischenko reaction is a modification of:

Online 2014 SET (2)

- (a) Pinacol-pinacolone reaction
 (b) Cannizzaro reaction
 (c) Claisen condensation
 (d) Aldol condensation

3. Chlorobenzene reacts with trichloro acetaldehyde in the presence of H_2SO_4

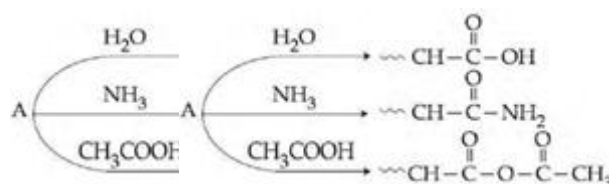
Online 2014 SET (3)



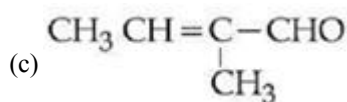
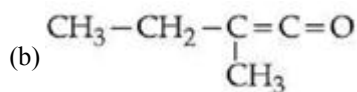
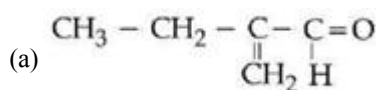
The major product formed is:

- (a)
- (b)
- (c)
- (d)

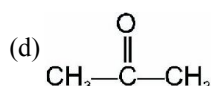
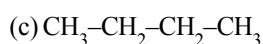
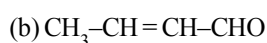
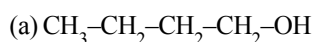
4. An organic compound A, $\text{C}_5\text{H}_8\text{O}$; reacts with H_2O , NH_3 and CH_3COOH as described below: Online 2014 SET (3)



A is:

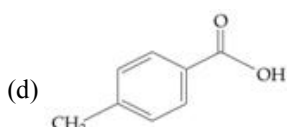
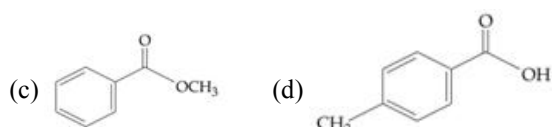
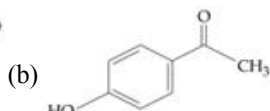
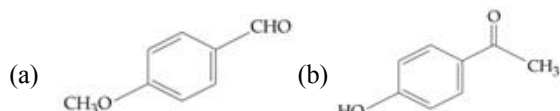


5. In the reaction sequence **Online 2015 SET (1)**

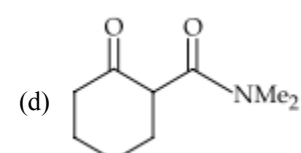
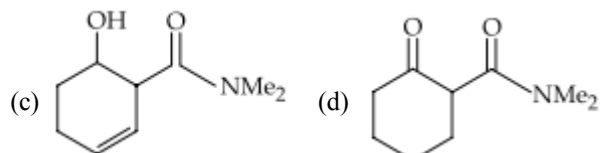
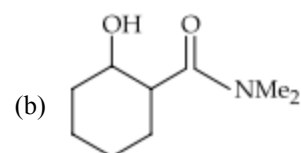
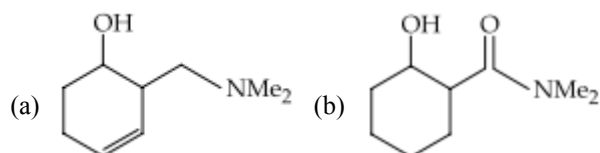
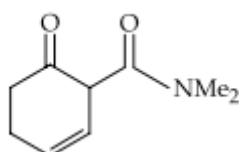


6. A compound of molecular formula $\text{C}_8\text{H}_8\text{O}_2$ reacts with acetophenone to form a single cross-aldol product in the presence of base. The same compound on reaction with conc. NaOH forms benzyl alcohol as one of the products. The structure of the compound is :

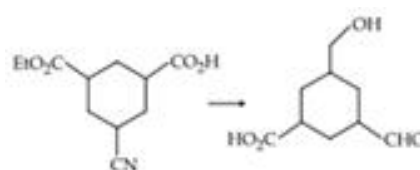
Online 2017 SET (2)



7. The main reduction product of the following compound with NaBH_4 in methanol is : **Online 2018 SET (1)**

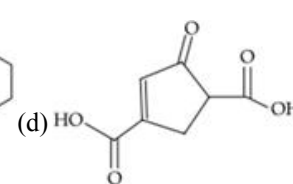
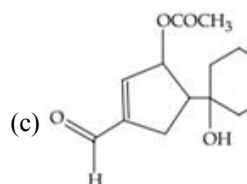
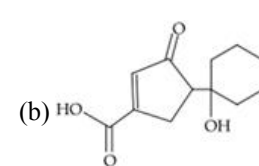
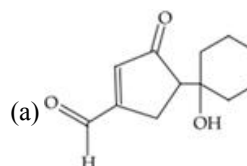
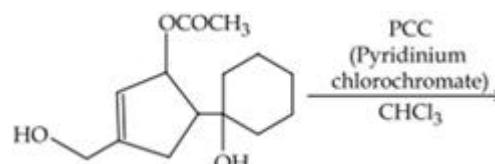


8. The reagent(s) required for the following conversion are : **Online 2018 SET (2)**



9. The major product formed in the following reaction is :

(Online 2019 SET 2)



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

- All questions marked "S" are single choice questions
- All questions marked "M" are multiple choice questions
- All questions marked "C" are comprehension based questions
- All questions marked "A" are assertion–reason type questions

(A) If both assertion and reason are correct and reason is the correct explanation of assertion.

(B) If both assertion and reason are true but reason is not the correct explanation of assertion.

(C) If assertion is true but reason is false.

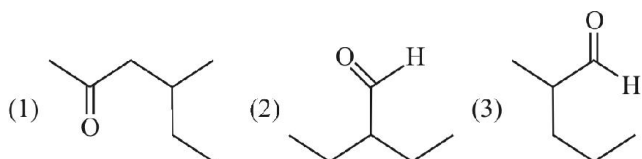
(D) If reason is true but assertion is false.
- All questions marked "X" are matrix–match type questions
- All questions marked "I" are integer type questions

Basic Concepts

- (S) Which of the following compound is resistant to nucleophilic attack by hydroxyl ions

(a) Methyl acetate (b) Acetonitrile

(c) Dimethyl ether (d) Acetamide
- (S) Which of the compound shown is (are) correctly named as pentane derivatives, either as pentanals or pentanones ?



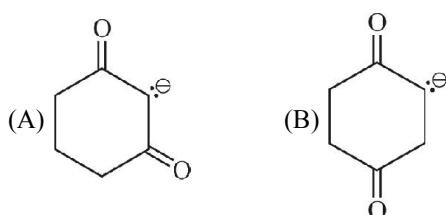
- (a) 1 only (b) 2 only
- (c) 3 only (d) 1 and 3
- (S) When a nucleophile encounters a ketone, the site of attack is

(a) The carbon atom of the carbonyl

(b) The oxygen atom of the carbonyl

(c) Both the carbon and oxygen atoms, with equal probability

(d) No attack occurs ketones do not react with nucleophiles.
 - (S) When enolate A is compared with enolate B



which of the following statements is true ?

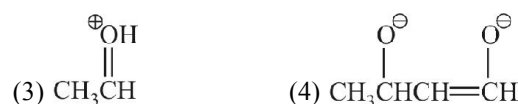
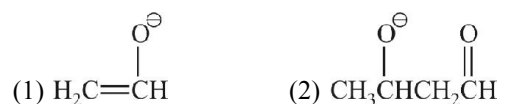
- (a) A is more stable than B
- (b) B is more stable than A
- (c) A and B have the same stability
- (d) No comparison of stability can be made.
- (S) Which of the following forms an enol to the greatest extent ?

(a) $\text{CH}_3\text{CH}_2\text{CHO}$

(b) $\text{CH}_3\text{C}(=\text{O})\text{CH}_2\text{CH}_2\text{C}(=\text{O})\text{CH}_3$

(c) $\text{CH}_3\text{C}(=\text{O})\text{CH}_2\text{C}(=\text{O})\text{CH}_3$

(d) $\text{CH}_3\text{C}(=\text{O})\text{C}(=\text{O})\text{CH}_2\text{CH}_3$
 - (S) Which of the following species is (are) not intermediates in the aldol condensation of acetaldehyde (ethanal) in aqueous base ?



- (a) 1 and 2
- (b) 3 only
- (c) 4 only
- (d) 3 and 4

7. (A) **Assertion** : Lower aldehydes and ketones are soluble in water but the solubility decreases as the molecular mass increases.

Reason : Distinction between aldehydes and ketones can be made by tollen's reagent.

- (a) A (b) B
(c) C (d) D

Preparation

8. (S) The product(s) obtained via oxymercuration ($\text{HgSO}_4 + \text{H}_2\text{SO}_4$) of 1-butyne would be

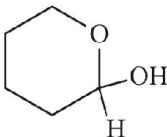
- (a) $\text{CH}_3\text{CH}_2\text{COCH}_3$ (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
(c) $\text{CH}_3\text{CH}_2\text{CHO} + \text{HCHO}$
(d) $\text{CH}_3\text{CH}_2\text{COOH} + \text{HCOOH}$

9. (A) **Assertion** : Reimer-Tiemann reaction is used to prepare benzaldehyde from benzene.

Reason : Reimer-Tiemann reaction involves electrophilic substitution.

- (a) A (b) B
(c) C (d) D

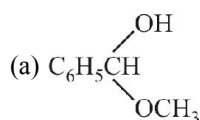
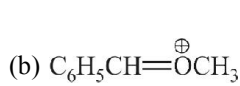
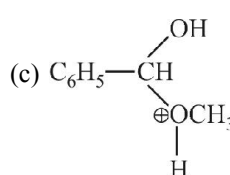
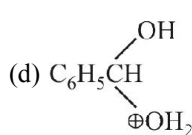
Chemical Reactions

10. (S) The structure  would be best classified

as a(n)

- (a) Acetal (b) Hemiacetal
(c) Hydrate (d) Cyanohydrin

11. (S) Which of the following species is the conjugate acid of the hemiacetal formed by reaction of benzaldehyde with methanol containing a trace of acid ?

- (a)  (b) 
(c)  (d) 

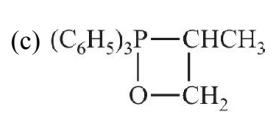
12. (S) A compound has a vapour density of 29. On warming an aqueous solution of alkali, it gives a yellow precipitate. The compound is

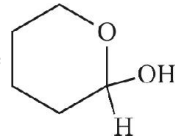
- (a) $\text{CH}_3\text{CH}_2\text{CHO}$ (b) $\text{CH}_3\text{CHOHCH}_3$
(c) CH_3COCH_3 (d) $\text{CH}_3\text{CH}_2\text{COOH}$

13. (S) Which is not true about acetophenone

- (a) Reacts to form 2, 4-dinitrophenyl hydrazine
(b) Reacts with Tollen's reagent to form silver mirror
(c) Reacts with I_2/NaOH to form iodoform
(d) On oxidation with alkaline KMnO_4 followed by hydrolysis gives benzoic acid

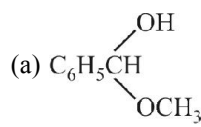
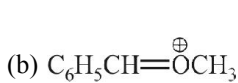
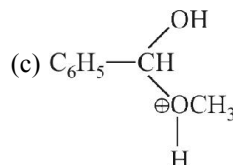
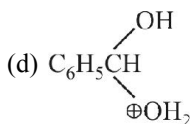
14. (S) Which of the following species is an ylide ?

- (a) $(\text{C}_6\text{H}_5)_3\text{P}^+\text{CH}_2\text{CH}_3\text{Br}^-$
(b) $(\text{C}_6\text{H}_5)_3\text{P}^+\text{CH}^-\text{CH}_3$
(c) 
(d) None of these

15. (S) The structure  would be best classified as a(n)

- (a) Acetal (b) Hemiacetal
(c) Hydrate (d) Cyanohydrin

16. (S) Which of the following species is the conjugate acid of the hemiacetal formed by reaction of benzaldehyde with methanol containing a trace of acid ?

- (a)  (b) 
(c)  (d) 

17. (A) **Assertion :** Aldehydes form stable hydrates.

Reason : Aldehydic carbonyl is more electron deficient than the ketone.

- (a) A (b) B
(c) C (d) D

Reduction Reactions

18. (S) Clemmenson reduction involves $>C=O$ to $>CH_2$ in presence of

- (a) Zn/Hg (b) Alcohol
(c) Zn dust (d) Zn/alcohol

19. (S) Clemenson's reduction of ketones is carried out in

- (a) H_2 with Pd catalyst (b) Glycol with KOH
(c) $LiAlH_4$ in water (d) Zn-Hg with HCl

20. (S) Aldehydes and ketones can be reduced to corresponding hydrocarbons by

- (a) Refluxing with water
(b) Refluxing with strong acids
(c) Refluxing with soda amalgam and water
(d) Refluxing with zinc amalgam and concentrated HCl

21. (S) In Wolf-Kishner reduction, the carbonyl group of aldehydes and ketones is converted into

- (a) $>CH_2$ group (b) $-CH_3$ group
(c) $-CH_2OH$ group (d) $>CHOH$ group

Conensation Reactions

22. (S) A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives

- (a) Benzyl alcohol and sodium formate
(b) Sodium benzoate and methyl alcohol
(c) Sodium benzoate and sodium formate
(d) Benzyl alcohol and methyl alcohol

23. (S) Aldol condensation will not take place in

- (a) HCHO (b) CH_3CH_2CHO
(c) CH_3CHO (d) CH_3COCH_3

24. (S) Aldol condensation involving $CH_3CHO + CH_3CHO$ gives the product

- (a) $CH_3CHOHCH_2CHO$ (b) $CH_3COCH_2CH_3$
(c) $CH_3CH=CH_2$ (d) None of these

25. (S) Cannizzaro reaction is given by

- (a) HCHO (b) CH_3COCH_3
(c) CH_3CHO (d) CH_3CH_2OH

26. (S) Cannizzaro reaction is not shown by

- (a) HCHO (b) C_6H_5CHO
(c) CH_3CHO (d) All of these

27. (S) Which one of the following undergoes reaction with 50% sodium hydroxide solution to give the corresponding alcohol and acid

- (a) Butanal (b) Benzaldehyde
(c) Phenol (d) Benzoic acid

28. (S) $OCH-CHO \xrightarrow{OH^-} HOH_2C-COOH$

The reaction given is

- (a) Aldol condensation (b) Knoevenagel reaction
(c) Cannizzaro reaction (d) None of these

29. (S) Which one does not give cannizzaro's reaction

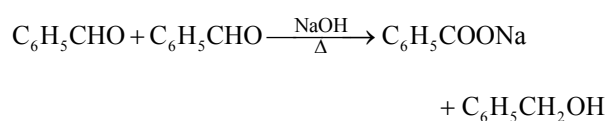
- (a) Benzaldehyde
(b) 2-methyl propanal
(c) p-methoxy benzaldehyde
(d) 2, 2-dimethyl propanal

30. (S) The key step in cannizzaro's reaction is the intermolecular shift of

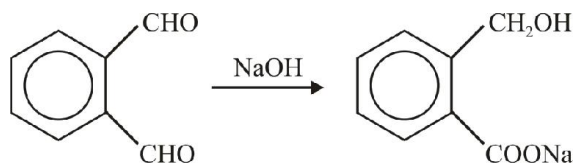
- (a) Proton (b) Hydride ion
(c) Hydronium ion (d) Hydrogen bond

Comprehension

Aldehydes undergo disproportionation reaction in presence of aqueous NaOH. Simultaneous oxidation and reduction of a compound is scientifically called as disproportionation. Aldehydes having no α -hydrogen show this reaction called Cannizzaro's reaction. Few exceptions are also there to this generalisation.



Intramolecular Cannizzaro's reaction is also possible.



Answer the following questions :

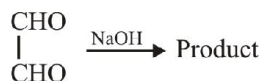
31. (C) The aldehyde which shows Cannizzaro's reaction is

- (a) HCHO (b) $\text{C}_6\text{H}_5\text{CHO}$
(c) $\text{CCl}_3\text{-CHO}$ (d) All of these

32. (C) The aldehyde having α -hydrogen which gives cannizzaro's reaction is :

- (a) $\text{CH}_3\text{-CH}_2\text{CHO}$ (b) $\text{CH}_3\text{-CH(CH}_3\text{)-CHO}$
(c) $\text{CH}_3\text{-CH(Cl)-CHO}$ (d) $\text{C}_2\text{H}_5\text{-CH}_2\text{-CHO}$

33. (C) The product formed in the following reaction will be :



- (a) CH_2OH
|
 CH_2OH (b) CH_2OH
|
 COONa
(c) COONa
|
 COONa (d) all of these

34. (C) Mixture of $\text{C}_6\text{H}_5\text{CHO}$ and HCHO is treated with NaOH then Cannizzaro's reaction involves :

- (a) Oxidation of HCHO
(b) Reduction of HCHO
(c) Oxidation of $\text{C}_6\text{H}_5\text{CHO}$
(d) Reduction of $\text{C}_6\text{H}_5\text{CHO}$

35. (C) Which of the following compounds gives internal crossed Cannizzaro's reaction ?

- (a) $\text{C}_6\text{H}_5\text{-CH}_2\text{-C(=O)-H}$ (b) $\text{C}_6\text{H}_5\text{-C(=O)-CHO}$
(c) (d)

Chemical Tests

36. (S) Which of the following does not give brick red precipitate with Fehling solution

- (a) Acetone (b) Acetaldehyde
(c) Formalin (d) D-glucose

37. (S) Acetaldehyde and acetone can be distinguished by

- (a) Molisch test (b) Bromoform test
(c) Solubility in water (d) Tollen's test

38. (S) The reagent that gives an orange coloured precipitate with acetaldehyde

- (a) NH_2OH (b) NaHSO_3
(c) Iodine (d) 2, 4-DNP

39. (S) The reagent with which both aldehyde and acetone react easily is

- (a) Fehling's reagent (b) Grignard reagent
(c) Schiff's reagent (d) Tollen's reagent

40. (S) Which compound undergoes iodoform reaction

- (a) HCHO (b) CH_3CHO
(c) CH_3OH (d) CH_3COOH

41. (S) Acetone and acetaldehyde are differentiated by

- (a) $\text{NaOH} + \text{I}_2$ (b) $\text{Ag}(\text{NH}_3)_2^+$
(c) HNO_2 (d) I_2

42. (S) Which of the following will not give iodoform test ?

- (a) Isopropyl alcohol (b) Ethanol
(c) Ethanal (d) Benzyl alcohol

Mixed Concepts

43. (X) Match the following :

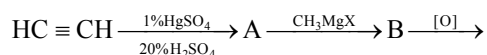
- | | |
|----------------------------|--|
| (A) Formaldehyde | (1) Iodoform test |
| (B) Acetone | (2) Test of aldehydes |
| (C) Acetaldehyde | (3) Ammonical silver nitrate |
| (D) Haloform reaction | (4) Polymer of phenol and formaldehyde |
| (E) Schiff's reagent | (5) Calcium acetate |
| (F) Clemmensen's reduction | (6) Urotropine |
| (G) Bakelite | (7) Boiling point 21°C |
| (H) Tollen's reagent | (8) Zinc amalgam and conc. HCl |

44. (X) Match the following

- | | |
|----------------------------|---|
| (A) Cannizzaro's reaction | (1) $\text{CH}_2=\text{CH}_2 + \text{PdCl}_2 + \text{H}_2\text{O} \xrightarrow{\text{CuCl}_2} \text{CH}_3\text{CHO} + \text{Pd} + 2\text{HCl}$ |
| (B) Rosenmund's reaction | (2) $(\text{CH}_3)_2\text{CO} + \text{H} \cdot \text{CH}_2\text{COCH}_3 \xrightarrow{\text{Ba(OH)}_2} (\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{COCH}_3$ |
| (C) Oxo process | (3) $\text{Zn/Hg} + \text{Conc. HCl}$ |
| (D) Wacker process | (4) $2\text{HCHO} \xrightarrow{\text{NaOH}} \text{HCOONa} + \text{CH}_3\text{OH}$ |
| (E) Clemmensen's reduction | (5) $\text{H}_2/\text{Pd} - \text{BaSO}_4$ |
| (F) Aldol condensation | (6) $(\text{C}_2\text{H}_5\text{O})_3\text{Al}$ |
| (G) Tischenko's reaction | (7) $\text{Alkene} + \text{CO} + \text{H}_2 \xrightarrow{\text{CO}_2(\text{CO})_8} \text{Aldehyde}$ |
| (H) Fehling's solution | (8) Red ppt. of Cu_2O |

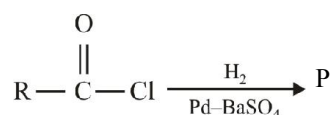
Preparation

45. (S) The end product in the following sequence of reaction is



- | | |
|-----------------|-----------------------|
| (a) Acetic acid | (b) Isopropyl alcohol |
| (c) Acetone | (d) Ethanol |

46. (S) In the following reaction, product P is



- | | |
|-----------------------------|--------------------|
| (a) RCH_2OH | (b) RCOOH |
| (c) RCHO | (d) RCH_3 |

47. (S) Acetophenone is prepared from

- | |
|----------------------------|
| (a) Rosenmund reaction |
| (b) Sandmeyer reaction |
| (c) Wurtz reaction |
| (d) Friedel craft reaction |

48. (S) Compound which gives acetone on ozonolysis

- | |
|---|
| (a) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ |
| (b) $(\text{CH}_3)_2\text{C} = \text{C}(\text{CH}_3)_2$ |
| (c) $\text{C}_6\text{H}_5\text{CH} = \text{CH}_2$ |
| (d) $\text{CH}_3\text{CH} = \text{CH}_2$ |

49. (S) Which one of the following compounds is prepared in the laboratory from benzene by a substitution reaction

- | | |
|------------------|---------------------------|
| (a) Glyoxal | (b) Cyclohexane |
| (c) Acetophenone | (d) Hexabromo cyclohexane |

50. (S) The oxidation of toluene to benzaldehyde by chromyl chloride is called

- | |
|-----------------------------|
| (a) Cannizzaro reaction |
| (b) Wurtz reaction |
| (c) Etard reaction |
| (d) Reimer-Tiemann reaction |

51. (S) $\text{CH}_3 - \text{CH}_2 - \text{C} \equiv \text{CH} \xrightarrow[\text{H}_2\text{O}]{\text{R}}$ Butanone, R is

- | | |
|----------------------|---------------------------------------|
| (a) Hg^{++} | (b) KMnO_4 |
| (c) KClO_3 | (d) $\text{K}_2\text{Cr}_2\text{O}_7$ |

52. (S) Which of the following gases when passed through warm dilute solution of H_2SO_4 in presence of HgSO_4 gives acetaldehyde

- | | |
|----------------------------|----------------------------|
| (a) CH_4 | (b) C_2H_6 |
| (c) C_2H_4 | (d) C_2H_2 |

Chemical Reactions

53. (S) $\text{CH}_3\text{CHO} + \text{CH}_3\text{MgBr} \rightarrow \text{Product} \xrightarrow{\text{H}_2\text{O}}$ A. What is A ?

- (a) Primary alcohol (b) Secondary alcohol
(c) Tertiary alcohol (d) Ketone

54. (S) Product in following reaction is $\text{CH}_3\text{MgI} + \text{HCHO} \rightarrow$ Product

- (a) CH_3CHO (b) CH_3OH
(c) $\text{C}_2\text{H}_5\text{OH}$ (d) $\text{CH}_3-\text{O}-\text{CH}_3$

55. (S) A compound has a vapour density of 29. On warming an aqueous solution of alkali, it gives a yellow precipitate. The compound is

- (a) $\text{CH}_3\text{CH}_2\text{CHO}$ (b) $\text{CH}_3\text{CHOHCH}_3$
(c) CH_3COCH_3 (d) $\text{CH}_3\text{CH}_2\text{COOH}$

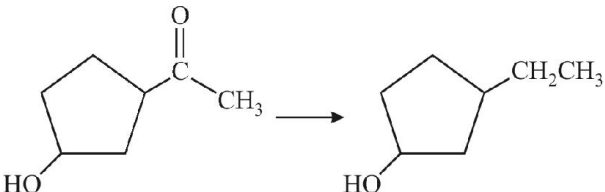
56. (S) Which is not true about acetophenone

- (a) Reacts to form 2, 4-dinitrophenyl hydrazine
(b) Reacts with Tollen's reagent to form silver mirror
(c) Reacts with I_2/NaOH to form iodoform
(d) On oxidation with alkaline KMnO_4 followed by hydrolysis gives benzoic acid

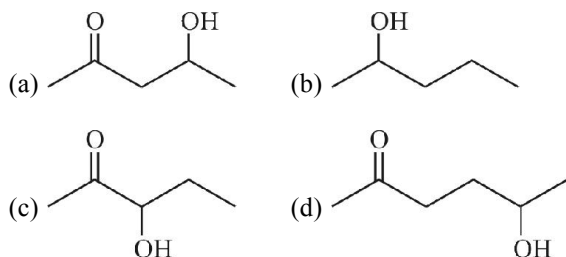
EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

Single Answer Questions

1. The reagent with which both acetaldehyde and acetone react easily, is (1981)
(a) Tollen's reagent (b) Schiff's reagent
(c) Grignard reagent (d) Fehling reagent
2. A compound that gives a positive iodoform test is (1982)
(a) 1-pentanol (b) 3-pentanone
(c) 2-pentanone (d) pentanal
3. When acetaldehyde is heated with Fehling's solution, it gives a precipitate of (1983)
(a) Cu (b) CuO
(c) Cu₂O (d) Cu + Cu₂O + CuO
4. The Cannizaro's reaction is not given by (1984)
(a) trimethyl acetaldehyde (b) acetaldehyde
(c) benzaldehyde (d) formaldehyde
5. The compound that will not give iodoform on treatment with alkali and iodine, is (1985)
(a) acetone (b) ethanol
(c) diethyl ketone (d) isopropyl alcohol
6. Which of the following compounds is oxidised to prepare methyl ethyl ketone ? (1987)
(a) 2-propanol (b) 1-butanol
(c) 2-butanol (d) t-butyl alcohol
7. m-chlorobenzaldehyde on reaction with conc. KOH at room temperature gives (1991)
(a) potassium m-chlorobenzoate and m-chlorobenzyl alcohol
(b) m-hydroxy benzaldehyde and m-chlorobenzyl alcohol
(c) m-chlorobenzyl alcohol and m-hydroxy benzyl alcohol
(d) potassium m-chlorobenzoate and m-hydroxy benzaldehyde
8. In the Cannizaro's reaction given below : (1996)
$$2\text{Ph} - \text{CHO} \xrightarrow{\text{OH}^-} \text{Ph} - \text{CH}_2\text{OH} + \text{PhCO}_2^-$$

The slowest step is
(a) the attack of OH at the carbonyl group
(b) the transfer of hydride to the carbonyl group
(c) the abstraction of proton from the carboxylic acid
(d) the deprotonation of Ph-CH₂OH
9. Which of the following will react with water ? (1998)
(a) CHCl₃ (b) Cl₃C.CHO
(c) CCl₄ (d) ClCH₂CH₂Cl
10. The enol form of acetone, after treatment with D₂O, gives (1999)
(a) $\text{CH}_3 - \overset{\text{OD}}{\underset{|}{\text{C}}} = \text{CH}_2$ (b) $\text{CD}_3 - \overset{\text{O}}{\underset{||}{\text{C}}} - \text{CD}_3$
(c) $\text{CH}_2 = \overset{\text{OH}}{\underset{|}{\text{C}}} - \text{CH}_2\text{D}$ (d) $\text{CD}_3 = \overset{\text{OD}}{\underset{|}{\text{C}}} - \text{CD}_3$
11. Which of the following has the most acidic hydrogen ? (2000)
(a) 3-hexanone (b) 2, 4-hexanedione
(c) 2, 5-hexanedione (d) 2, 3-hexanedione
12. The appropriate reagent for the following transformation (2000)

is
(a) Zn(Hg), HCl (b) NH₂NH₂, OH⁻
(c) H₂/Ni (d) NaBH₄

13. Which one of the following will most readily be dehydrated in acidic condition ? (2000)

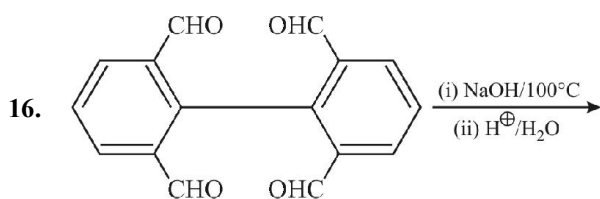


14. A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives (2001)

- (a) benzyl alcohol and sodium formate
(b) sodium benzoate and methyl alcohol
(c) sodium benzoate and sodium formate
(d) benzyl alcohol and methyl alcohol

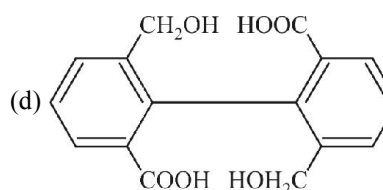
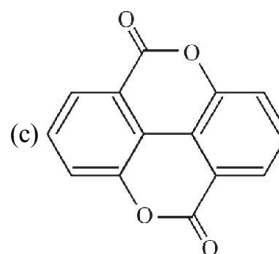
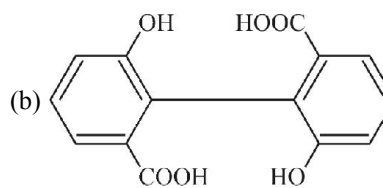
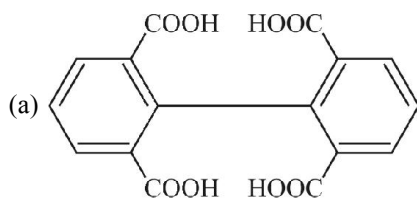
15. Compound A (molecular formula C_3H_8O) is treated with acidified potassium dichromate to form a product B (molecular formula C_3H_6O). B forms a shining silver mirror on warming with ammoniacal silver nitrate. B when treated with an aqueous solution of $H_2NCONHNH_2$ and sodium acetate gives a product C. Identify the structure of C. (2002)

- (a) $CH_3CH_2CH=N.NHCONH_2$
(b) $CH_3-C(=N.NHCONH_2)-CH_3$
(c) $CH_3-C(=N.CONHNH_2)-CH_3$
(d) $CH_3CH_2CH=N.CONHNH_2$

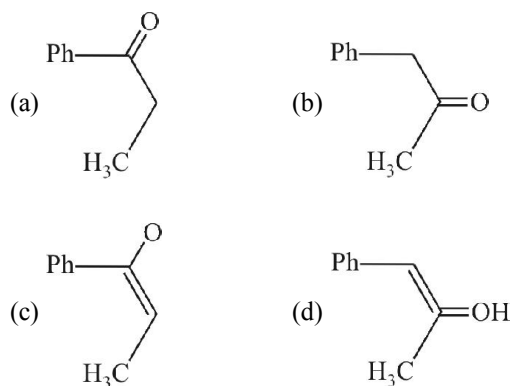


major product is

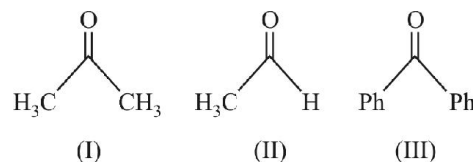
(2003)



17. $Ph-C\equiv C-CH_3 \xrightarrow{Hg^{2+}/H^+} A$, A is (2003)



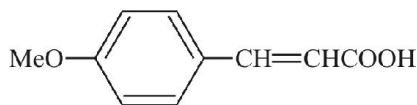
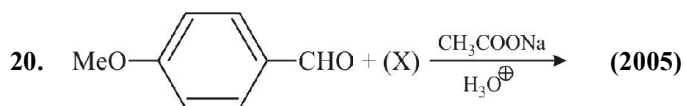
18. The order of reactivity of phenyl magnesium bromide with the following compounds is (2004)



- (a) (II) > (III) > (I) (b) (I) > (III) > (II)
(c) (II) > (I) > (III) (d) All react with the same rate

19. How will you convert butan-2-one to propanoic acid ? (2005)

- (a) Tollen's reagent (b) Fehling solution
(c) $NaOH/I_2/H^+$ (d) $NaOH/NaI/H^+$



the compound (X) is

- (a) CH_3COOH (b) BrCH_2COOH
(c) $(\text{CH}_3\text{CO})_2\text{O}$ (d) $\text{CHO}-\text{COOH}$

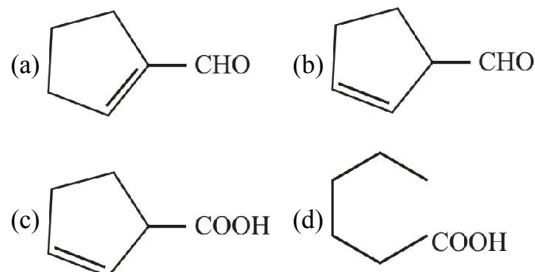
21. Butan-2-one can be converted to propanoic acid by which of the following? (2006)

- (a) NaOH , NaI/H^+ (b) Fehling solution
(c) NaOH , I_2/H^+ (d) Tollen's reagent

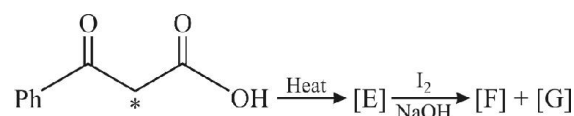
22. The smallest ketone and its next homologue are reacted with NH_2OH to form oxime (2006)

- (a) two different oximes are formed
(b) three different oximes are formed
(c) two oximes are optically active
(d) all oximes are optically active

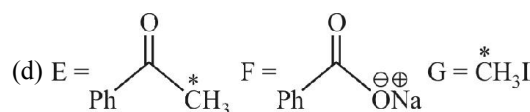
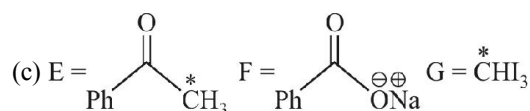
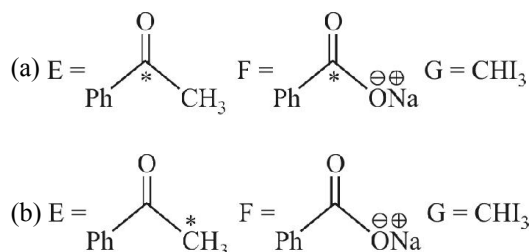
23. Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound E. Compound E on further treatment with aqueous KOH yields compound F. Compound F is (2007)



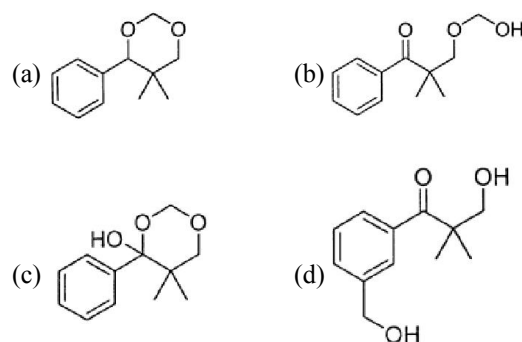
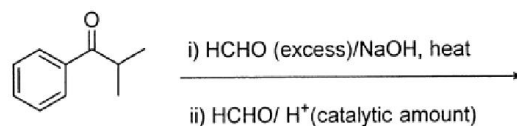
24. In the following reaction sequence, the correct structures of E, F and G are (2008)



(* implies ^{13}C labelled carbon)



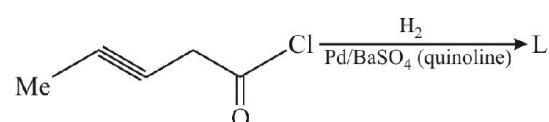
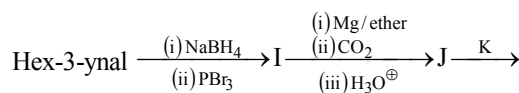
25. The major product of the following reaction sequence is (2012)



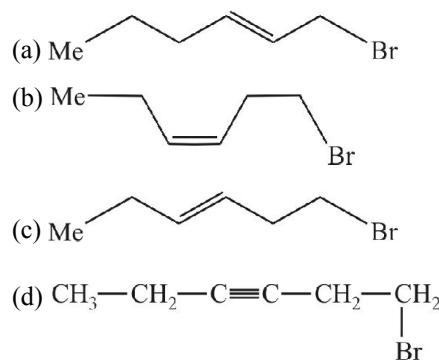
Comprehension

Passage – 1 (Ques. 26 to 28)

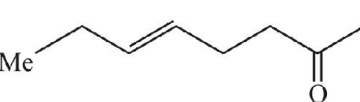
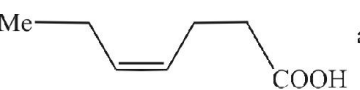
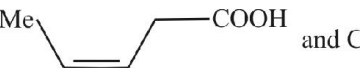
In the following reaction sequence, product I, J and L are formed. K represents a reagent. (2008)



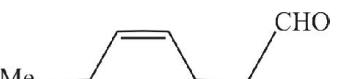
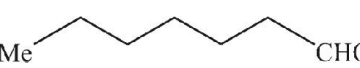
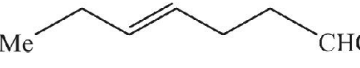
26. The structure of the product I is (2008)



27. The structures of compounds J and K, respectively, are (2008)

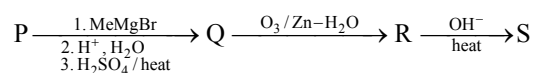
- (a)  and SOCl_2
- (b)  and SO_2Cl_2
- (c)  and SOCl_2
- (d)  and $\text{CH}_3\text{SO}_2\text{Cl}$

28. The structure of product L is (2008)

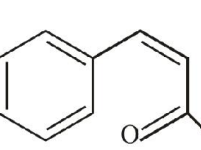
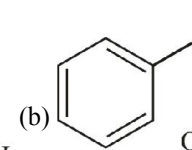
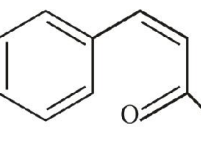
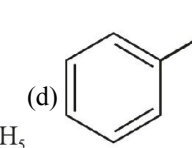
- (a) 
- (b) 
- (c) 
- (d) 

Passage – 2 (Ques. 29 to 31)

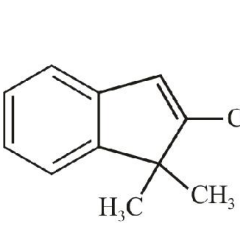
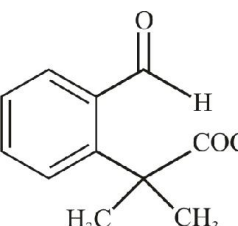
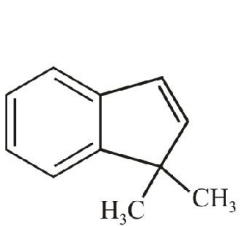
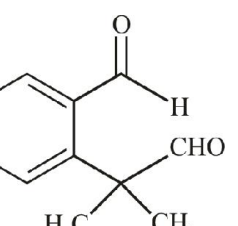
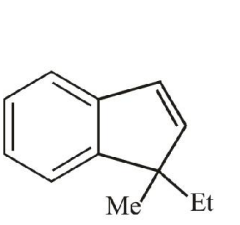
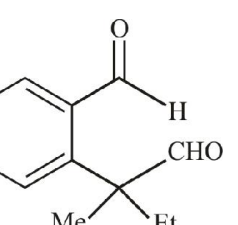
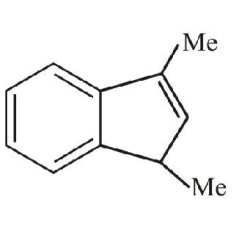
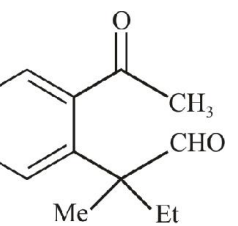
A carbonyl compound P, which gives positive iodoform test, undergoes reaction with MeMgBr followed by dehydration to give an olefin Q. Ozonolysis of Q leads to a dicarbonyl compound R, which undergoes intramolecular aldol reaction to give predominantly S.



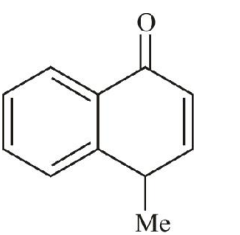
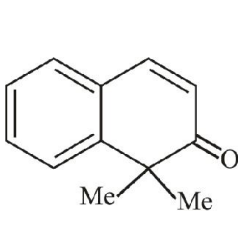
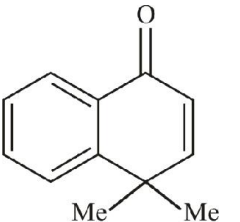
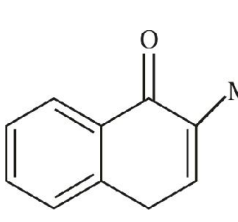
29. The structure of the carbonyl compound P, is (2010)

- (a) 
- (b) 
- (c) 
- (d) 

30. The structures of the products Q and R, respectively, are (2010)

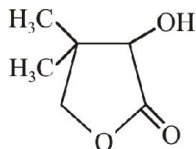
- (a)  
- (b)  
- (c)  
- (d)  

31. The structure of the product S, is (2010)

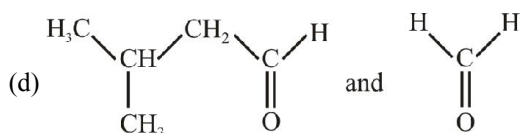
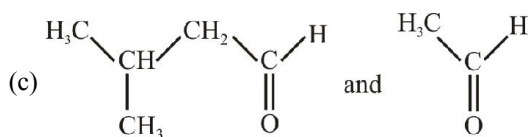
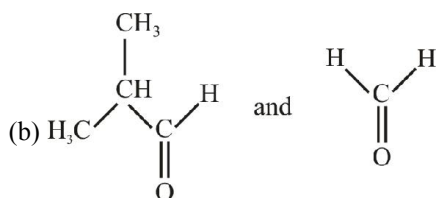
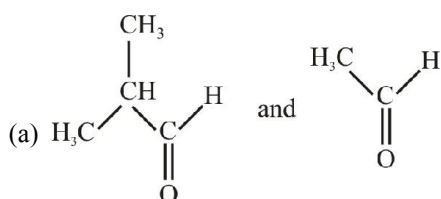
- (a) 
- (b) 
- (c) 
- (d) 

Passage – 3 (Ques. 32 to 34)

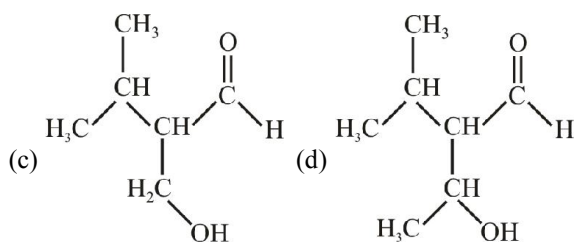
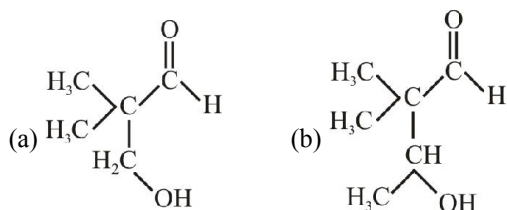
Two aliphatic aldehydes P and Q react in the presence of aqueous K_2CO_3 to give compound R, which upon treatment with HCN provides compound S. On acidification and heating, S gives the product shown below :



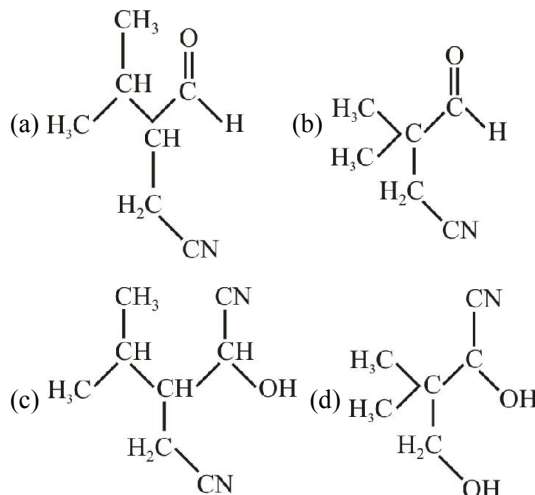
32. The compounds P and Q respectively are (2011)



33. The compound R is (2011)



34. The compound S is (2011)


Match the Columns

35. Match the compounds/ions in Column-I with their properties/reactions in Column-II. (2007)

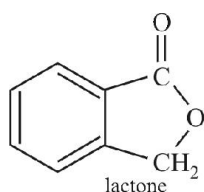
Column-I	Column-II
(A) C_6H_5CHO	(P) gives precipitate with 2, 4-dinitrophenylhydrazine
(B) $CH_3C\equiv CH$	(Q) gives precipitate with $AgNO_3$
(C) $CN^\ominus$	(R) is a nucleophile
(D) $I^\ominus$	(S) is involved in cyanohydrin formation

36. Match each of the compounds given in column I with the reaction (s) that they can undergo, given in column II. (2014)

Column I	Column II
(A)	(P) Nucleophilic substitution
(B)	(Q) Elimination
(c)	(R) Nucleophilic addition
(D)	(S) Esterification with acetic anhydride
	(T) Dehydrogenation

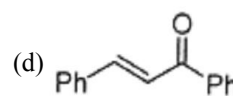
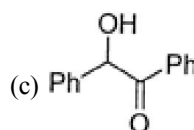
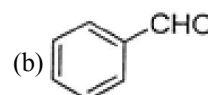
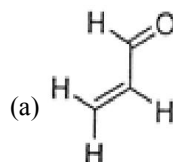
Multiple Answer Questions

37. Base catalysed aldol condensation occurs with (1984)
(a) propionaldehyde (b) benzaldehyde
(c) 2-methyl propionaldehyde
(d) 2, 2-dimethyl propionaldehyde
38. Which of the following compounds will give a yellow precipitate with iodine and alkali? (1984)
(a) 2-hydroxy propane (b) Acetophenone
(c) Methyl acetate (d) Acetamide
39. Which of the following compounds will react with ethanolic KCN? (1984)
(a) Ethyl chloride (b) Acetyl chloride
(c) Chlorobenzene (d) Benzaldehyde
40. Which of the following are the examples of aldol condensation? (1989)
(a) $2\text{CH}_3\text{CHO} \xrightarrow{\text{Dil. NaOH}} \text{CH}_3\text{CHOHCH}_2\text{CHO}$
(b) $2\text{CH}_3\text{COCH}_3 \xrightarrow{\text{Dil. NaOH}} \text{CH}_3\text{COHCH}_3\text{CH}_2\text{COCH}_3$
(c) $2\text{HCHO} \xrightarrow{\text{Dil. NaOH}} \text{CH}_3\text{OH}$
(d) $\text{C}_6\text{H}_5\text{CHO} + \text{HCHO} \xrightarrow{\text{Dil. NaOH}} \text{C}_6\text{H}_5\text{CH}_2\text{OH}$
41. A new carbon-carbon bond formation is possible in (1998)
(a) Cannizzaro's reaction (b) Friedel-Craft's reaction
(c) Clemmensen reduction (d) Reimer-Tiemann reaction
42. Among the following compounds, which will react with acetone to give a product containing $\text{C}=\text{N}$? (1998)
(a) $\text{C}_6\text{H}_5\text{NH}_2$ (b) $(\text{CH}_3)_3\text{N}$
(c) $\text{C}_6\text{H}_5\text{NHC}_6\text{H}_5$ (d) $\text{C}_6\text{H}_5\text{NHNH}_2$
43. Which of the following will undergo aldol condensation? (2003)
(a) Acetaldehyde (b) Propanaldehyde
(c) Benzaldehyde (d) Trideutero acetaldehyde
44. Which of the following reactants on reaction with conc. NaOH followed by acidification gives following lactone as the main product? (2008)

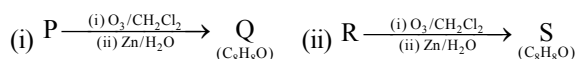


- (a) (b)
(c) (d)

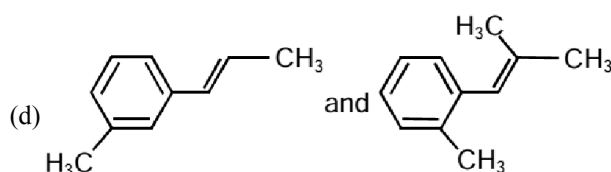
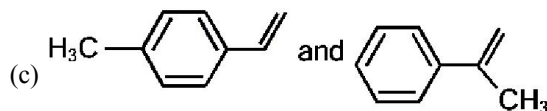
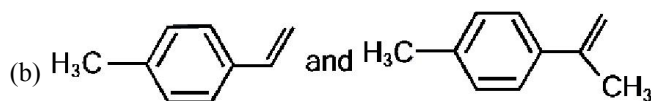
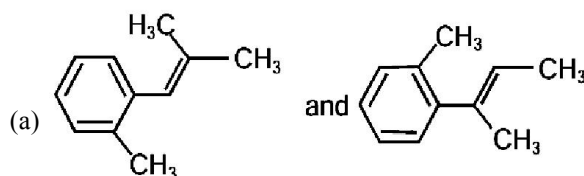
45. The smallest ketone and its next homologue are reacted with NH_2OH to form oxime (2010)
(a) two different oximes are formed
(b) three different oximes are formed
(c) two oximes are optically active
(d) all oximes are optically active
46. Positive Tollen's test is observed for (2016)



47. Compound P and R upon ozonolysis produce Q and S, respectively. The molecular formula of Q and S is $\text{C}_8\text{H}_8\text{O}$. Q undergoes Cannizzaro reaction but not haloform reaction, whereas S undergoes haloform reaction but not Cannizzaro reaction. (2017)



The option(s) with suitable combination of P and R, respectively, is (are)



Fill in the Blanks

48. Fehling's solution 'A' consists of an aqueous solution of copper sulphate, while Fehling's solution 'B' consists of an alkaline solution (1990)

True/False

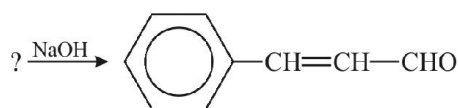
49. Benzaldehyde undergoes aldol condensation in an alkaline medium. (1982)
50. The yield of ketone when a secondary alcohol is oxidised is more than the yield of aldehyde when a primary alcohol is oxidised. (1983)
51. The reaction of methyl magnesium iodide with acetone followed by hydrolysis gives secondary butanol. (1987)

Subjective Questions

52. Write the structural formula of the main organic product formed when methanal reacts with ammonia. (1981)
53. Outline the reaction sequence for the conversion of methanal to ethanal (the number of steps should not be more than three). (1981)
54. An alkene 'A' on ozonolysis yields acetone and an aldehyde. The aldehyde is easily oxidised to an acid 'B'. When 'B' is treated with bromine in presence of phosphorus it yields a compound 'C' which on hydrolysis gives a hydroxy acid 'D'. This acid can also be obtained from acetone by the reaction with hydrogen cyanide followed by hydrolysis. Identify the compounds A, B, C and D. (1982)
55. Show with balanced equation what happens when the following are mixed : (1984)
"Chloral is heated with aqueous hydroxide."
56. Write down the reactions involved in the preparation of the following using the reagents indicated against in parenthesis : (1984)
"Acetoxime from acetaldehyde."
[$K_2Cr_2O_7/H^+$, $Ca(OH)_2$ and $NH_2OH.HCl$]
57. Arrange the following in order of their increasing reactivity towards HCN : (1985)
 CH_3CHO , CH_3COCH_3 , $HCHO$, $C_2H_5COCH_3$
58. Write down the main product of the following reaction (1985)
$$\text{Propanal} \xrightarrow[\text{heat}]{NaOH}$$
59. Give reasons in one or two sentences for the following (1986)
"Hydrazone of aldehydes and ketones are not prepared in highly acidic medium."

60. How may the following transformation be carried out (in not more than six steps) "Benzaldehyde to cyanobenzene" ? (1986)

61. Complete the following with appropriate structures (1986)



62. Answer the following with suitable equations wherever necessary (1987)

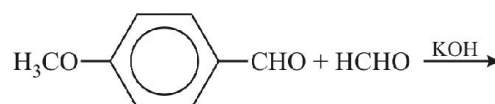
- (i) Suggest a reagent to distinguish acetaldehyde from acetone.
- (ii) What happens when excess chlorine is passed through boiling toluene in the presence of sun light ?

63. A ketone A, which undergoes haloform reaction, gives compound B on reduction. B on heating with sulphuric acid gives compound C, which forms mono-ozonide D. D on hydrolysis in presence of zinc dust gives only acetaldehyde. Identify A, B and C. Write down the reactions involved. (1989)

64. Arrange the following in increasing order of expected enol content. (1992)



65. Write the structure of the major organic product expected from the following reaction. (1992)

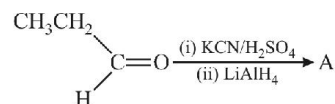


66. How will you bring about the following conversions ? (1994)
"4-nitrobenzaldehyde from benzene."

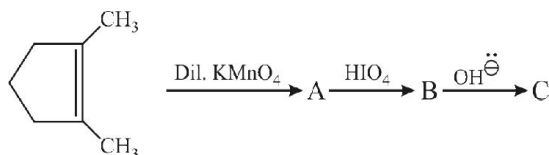
67. Complete the following reaction with appropriate structure. (1995)



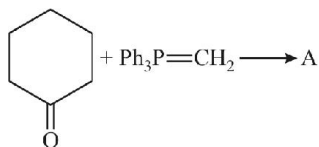
68. Complete the following reaction with appropriate structure (1996)



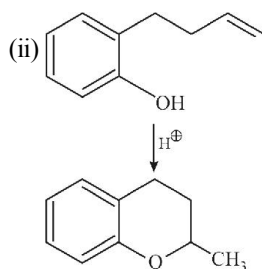
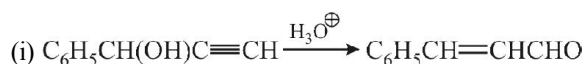
69. Suggest appropriate structures for the missing compounds. (The number of carbon atoms remains the same throughout the reaction) (1996)



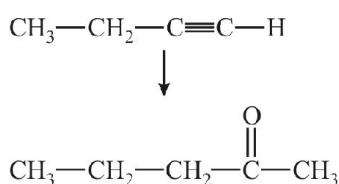
70. Complete the following, giving the structures of the principal organic products (1997)



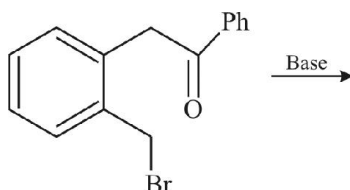
71. Acetophenone on reaction with hydroxyl amine hydrochloride can produce two isomeric oximes. Write structures of the oxime. (1997)
72. An aldehyde A ($C_{11}H_8O$), which does not undergo self aldol condensation, gives benzaldehyde and two moles of B on ozonolysis. Compound B, on oxidation with silver ion gives oxalic acid. Identify the compounds A and B. (1998)
73. Write the intermediate steps for each of the following reactions (1998)



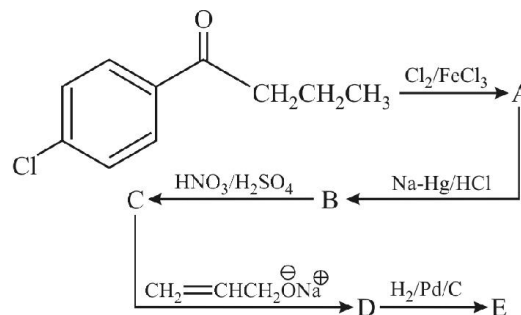
74. Explain, why o-hydroxy benzaldehyde is a liquid at room temperature while p-hydroxy benzaldehyde is a high melting solid? (1999)
75. Carry out the following transformation in not more than three steps (1999)



76. What would be the major product in the following reaction? (2000)



77. An organic compound A $C_6H_{10}O$, on reaction with CH_3MgBr followed by acid treatment gives compound B. The compound B on ozonolysis gives compound C, which in presence of a base gives 1-acetyl cyclopentene D. The compound B on reaction with HBr gives compound E. Write the structures of A, B, C, D and E. Show how D is formed from C. (2000)
78. An alkene (A) $C_{16}H_{16}$ on ozonolysis gives only one product (B) C_8H_8O . Compound (B) on reaction with $NaOH/I_2$ yields sodium benzoate. Compound (B) reacts with KOH/NH_2NH_2 yielding a hydrocarbon (C) C_8H_{10} . Write the structures of compounds (B) and (C). Based on this information two isomeric structures can be proposed for alkene (A). Write their structures and identify the isomer which on catalytic hydrogenation ($H_2/Pd-C$) gives a racemic mixture. (2001)
79. Write structures of the products A, B, C, D and E in the following scheme (2002)



80. A compound $C_9H_7O_2Cl$ exists predominantly in enol form (A) and also in keto form (B). On oxidation with $KMnO_4$ it gives m-chlorobenzoic acid as one of the products, Identify the compounds (A) and (B).
81. (A), $C_6H_{12} \xrightarrow{HCl}$ (B), $C_6H_{13}Cl$ + (C), $C_6H_{13}Cl$
 (B) $\xrightarrow{\text{alcoholic KOH}}$ (D) (an isomer of (A))
 (D) $\xrightarrow{\text{ozonolysis}}$ (E) (Positive iodoform and negative Fehling's solution test)
 (A) $\xrightarrow{\text{ozonolysis}}$ (F) + (G) (Positive Tollen's test for both)
 (F) + (G) $\xrightarrow{\text{conc. NaOH}}$ $HCOONa$ + A primary alcohol
 Identify the compounds (A) to (D).

ANSWER KEY

EXERCISE - 1 : (Basic Objective Questions)

1. (b)	2. (d)	3. (a)	4. (b)	5. (d)	6. (c)	7. (b)	8. (a)
9. (b)	10. (c)	11. (a)	12. (d)	13. (a)	14. (a)	15. (d)	16. (c)
17. (c)	18. (d)	19. (d)	20. (c)	21. (a)	22. (d)	23. (a)	24. (c)
25. (b)	26. (b)	27. (a)	28. (c)	29. (c)	30. (c)	31. (a)	32. (a)
33. (a)	34. (b)	35. (a)	36. (a)	37. (c)	38. (c)	39. (c)	40. (a)
41. (a)	42. (b)	43. (d)	44. (d)	45. (a)	46. (a)	47. (b)	48. (c)
49. (a)	50. (a)	51. (b)	52. (c)	53. (a)	54. (b)	55. (b)	56. (d)
57. (b)	58. (c)	59. (d)	60. (c)	61. (c)	62. (b)	63. (a)	64. (c)

EXERCISE - 2 : (Previous Year JEE Mains Questions)

1. (b)	2. (c)	3. (d)	4. (c)	5. (a)	6. (b)	7. (d)	8. (a)
9. (a)	10. (d)	11. (d)	12. (a)				

JEE Mains Online

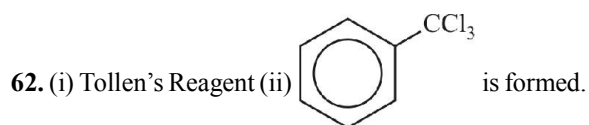
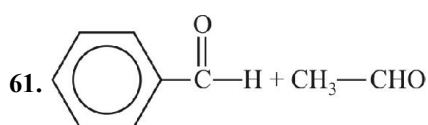
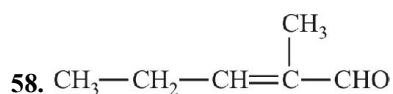
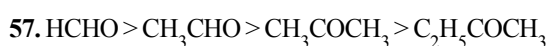
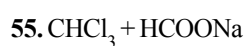
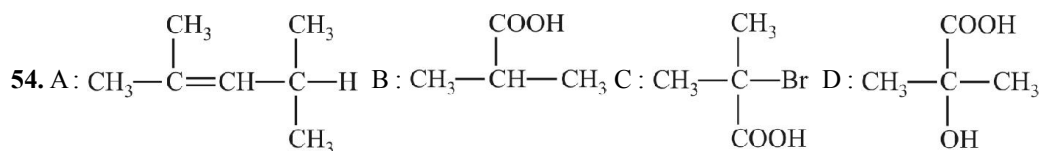
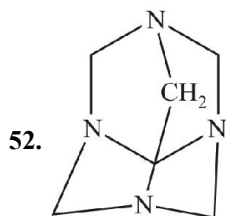
1. (c)	2. (b)	3. (c)	4. (b)	5. (b)	6. (a)	7. (c)	8. (c)
9. (c)							

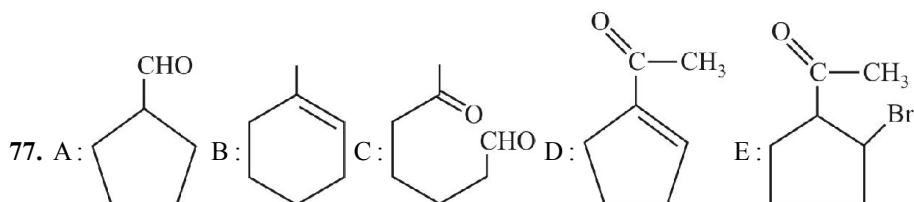
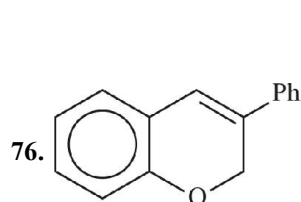
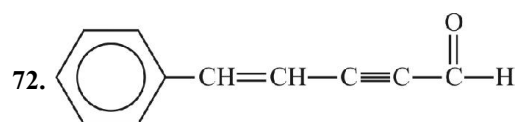
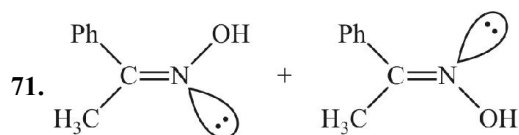
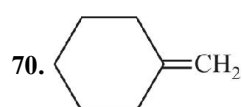
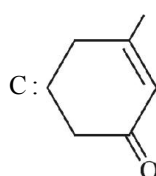
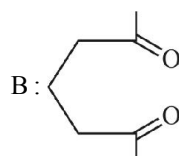
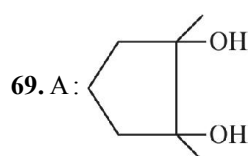
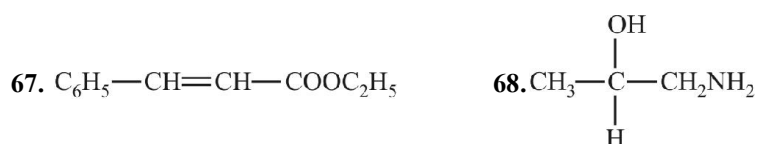
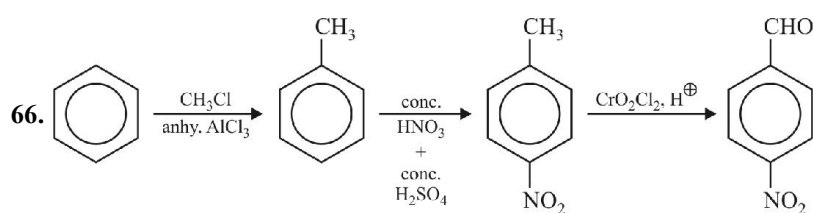
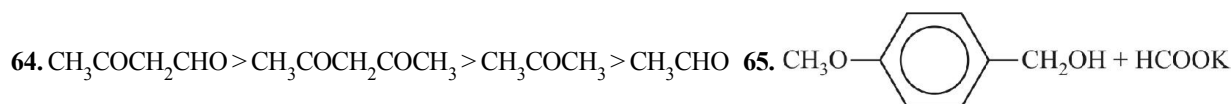
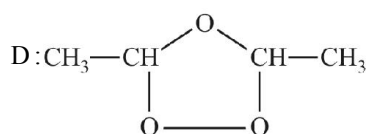
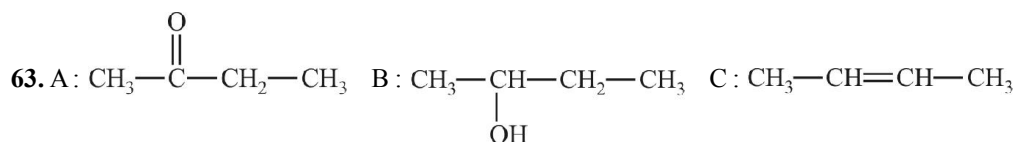
EXERCISE - 3 : (Advanced Objective Questions)

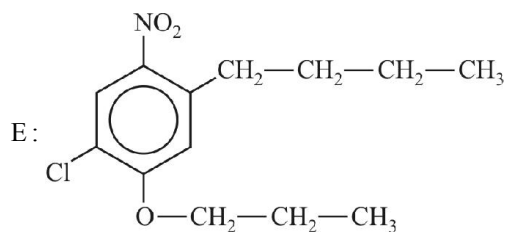
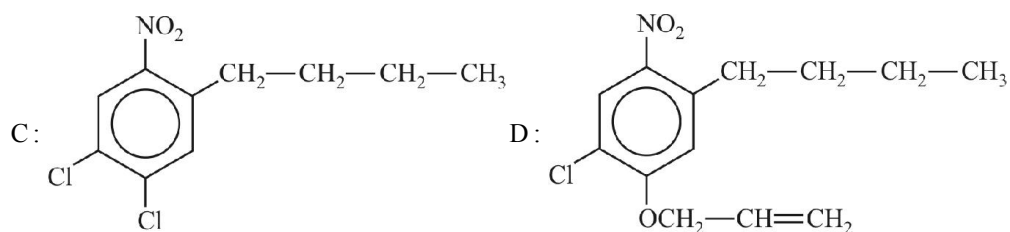
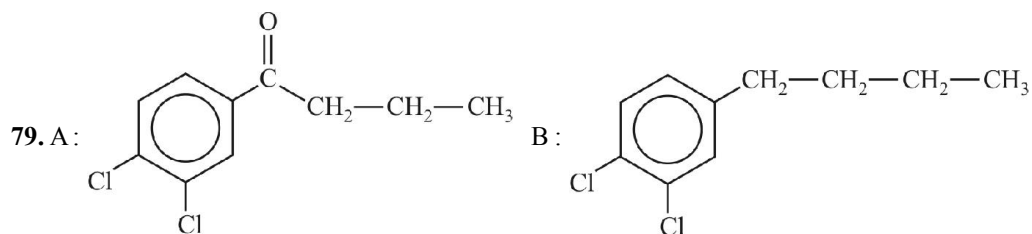
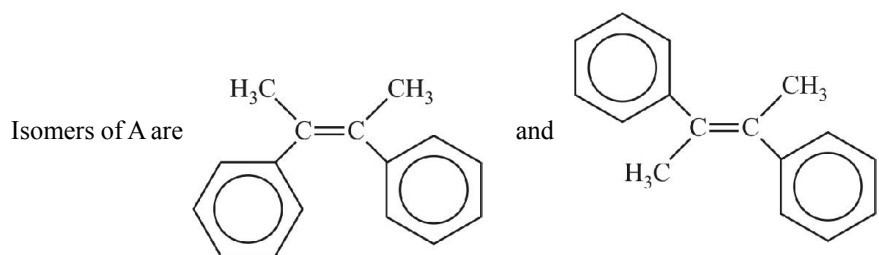
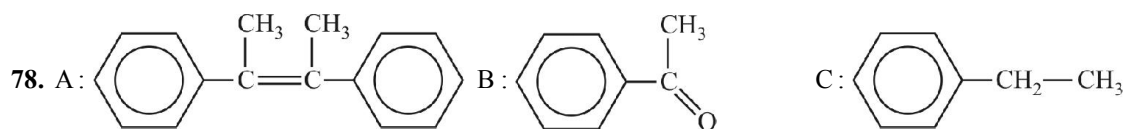
1. (c)	2. (c)	3. (a)	4. (a)	5. (c)	6. (d)	7. (b)	8. (a)
9. (d)	10. (b)	11. (c)	12. (c)	13. (b)	14. (b)	15. (b)	16. (c)
17. (d)	18. (a)	19. (d)	20. (d)	21. (a)	22. (a)	23. (a)	24. (a)
25. (a)	26. (c)	27. (b)	28. (c)	29. (b)	30. (b)	31. (d)	32. (b)
33. (b)	34. (a,d)	35. (b,c)	36. (a)	37. (d)	38. (d)	39. (b)	40. (b)
41. (b)	42. (d)	43. (A-6); (B-5); (C-7); (D-1); (E-2); (F-8); (G-4); (H-3)					
44. (A-4); (B-5); (C-7); (D-1); (E-3); (F-2); (G-6); (H-8)					45. (c)	46. (c)	47. (d)
48. (b)	49. (c)	50. (c)	51. (a)	52. (d)	53. (b)	54. (c)	55. (c)
56. (b)							

EXERCISE - 4 : (Previous Year JEE Advanced Questions)

1. (c) 2. (c) 3. (c) 4. (b) 5. (c) 6. (c) 7. (a) 8. (b)
9. (b) 10. (b) 11. (b) 12. (b) 13. (a) 14. (a) 15. (a) 16. (d)
17. (a) 18. (c) 19. (c) 20. (c) 21. (c) 22. (b) 23. (a) 24. (c)
25. (a) 26. (d) 27. (a) 28. (b) 29. (b) 30. (a) 31. (b) 32. (b)
33. (a) 34. (d) 35. (A) – P, Q, R, S ; (B) – Q, R ; (C) – R, S ; (D) – Q, R
36. A – P, Q, S ; B – Q, S ; C – Q, R, S ; D – Q, R 37. (a, c) 38. (a, b) 39. (a, d) 40. (a, b)
41. (b, d) 42. (d) 43. (a, b, d) 44. (b, c) 45. (b) 46. (a, b, c, d) 47. (b, c)
48. sodium potassium tartarate 49. (F) 50. (T) 51. (F)







Dream on !!

